

CYTOKINES

Human life within a narrow range: The lethal ups and downs of type I interferons

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The past 20 years have seen the definition of human monogenic disorders and their autoimmune phenocopies underlying either defective or enhanced type I interferon (IFN) activity. These disorders delineate the impact of type I IFNs in natural conditions and demonstrate that only a narrow window of type I IFN activity is beneficial. Insufficient type I IFN predisposes humans to life-threatening viral diseases (albeit unexpectedly few) with a central role in immunity to respiratory and cerebral viral infection. Excessive type I IFN, perhaps counterintuitively, appears to underlie a greater number of autoinflammatory and/or autoimmune conditions known as type I interferonopathies, whose study has revealed multiple molecular programs involved in the induction of type I IFN signaling. These observations suggest that the manipulation of type I IFN activity to within a physiological range may be clinically relevant for the prevention and treatment of viral and inflammatory diseases.

TYPE I IFNs IN ANTIVIRAL IMMUNITY

Type I IFNs are experimentally pleiotropic

Type I interferons (IFNs) were first described in 1957 as molecules secreted by virally infected nonleukocytic cells with strong antiviral activity in vitro on uninfected cells (1, 2). This discovery marked the birth of two fields: cytokine biology (as broadly defined by soluble molecules contributing to immunity via intercellular communication) and intrinsic immunity (as broadly defined by cell-intrinsic immunity against viruses in leukocytic or nonleukocytic cells). Type I IFNs are produced in all jawed vertebrates studied including humans and by most, if not all, nucleated cells. Particularly notable in this regard are plasmacytoid dendritic cells (pDCs), which produce greater quantities of type I IFNs than any other cell (3). By contrast, type II IFN, first discovered in leukocytes in 1965, exerts only modest antiviral activity. Rather, as first documented in 1983, type II IFN functions as a macrophage-activating factor. This function was confirmed from 1996 onward through the study of human inborn errors of type II IFN resulting in Mendelian susceptibility to mycobacterial disease (MSMD) and other intramacrophagic bacterial, fungal, or parasitic, but not viral, infections (4–6). The role of type I IFNs in antiviral immunity was soon documented in vivo in animal models, particularly in mice depleted of type I IFNs with specific antibodies and later in mice genetically null for the type I IFN receptor (7, 8). The type I IFNs were also found to exert antitumoral and antimicrobial (nonviral) activity in vitro and in animal models and a range of other pleiotropic activities across cell types, including in and between leukocytes (9, 10). The physiological importance of these nonviral activities in humans remains unclear (7, 11).

Type I IFNs are evolutionarily antiviral

Nearly all vertebrate cells studied can make one or more type I IFN, and nearly all cells can respond to type I IFNs (12). There are 17 type I IFNs in humans: 13 subtypes of IFN- α (encoded by *IFNA1*, *IFNA2*, *IFNA4*, *IFNA5*, *IFNA6*, *IFNA7*, *IFNA8*, *IFNA10*, *IFNA13*, *IFNA14*, *IFNA16*, *IFNA17*, and *IFNA21*) and IFN- β , IFN- ϵ , IFN- κ , and IFN- ω . However, the protein sequences for mature IFN- α 1 and IFN- α 13 are identical. Thus, the 13 *IFNA* genes encode 12 IFN- α proteins. The 16 human type I IFNs have different patterns of expression and function. IFN- β is the high-affinity short-lived priming IFN and the major type I IFN produced by fibroblasts, whereas IFN- ω and the 12 variants of IFN- α are produced in particularly large amounts by pDCs and circulate in the bloodstream. IFN- κ is expressed in the skin (13) and IFN- ϵ in the female reproductive tract (14).

These small proteins all bind the same heterodimeric receptor composed of interferon alpha and beta receptor subunit 1 (IFNAR1) and subunit 2 (IFNAR2). Studies of cellular responses to IFN led to the discovery of the Janus kinase–signal transducers and activators of transcription (JAK–STAT) pathway. The same JAK–STAT pathway—involving the ISGF-3 complex [composed of STAT1, STAT2, and IFN regulatory factor (IRF9)]—is also used by the type III IFNs, which are recognized by a single receptor comprising IFN lambda receptor-1 and interleukin-10 receptor 2 (IL-10R2) that is largely restricted to certain epithelial cell types. Type I and III IFNs are the only two multigenic families described to signal through a single receptor, which marks them out among cytokines, further suggesting that they operate as an antiviral network rather than as a set of discrete cytokines. All vertebrate species studied have multiple type I IFN genes, also indicating that the multilocus nature of type I IFNs has been evolutionarily driven by viral exposure.

Last, although fish type I IFN genes have introns, this is not true for all amphibian type I IFN genes. Moreover, all type I IFN genes in reptiles, birds, and mammals are intronless, consistent with an evolutionary pressure toward their rapid transcription on viral challenge (15–18). This inference was confirmed when humans genetically lacking functional type I IFN signaling were described. Such individuals exhibited no biological or clinical phenotype other than a susceptibility to viral infection. Thus, we now understand that the lifesaving, evolutionarily selected function of type I IFNs in natural

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conditions is to orchestrate an antiviral response, involving both nonleukocytic and leukocytic cell types. As discussed below, we now also appreciate that a glut of type I IFNs can be more detrimental in some senses than their deficiency, perhaps because the latter state only manifests in the context of a relevant environmental exposure (most particularly to certain viruses).

HUMAN INHERITED TYPE I IFN DEFICIENCIES

Deficits of the type I IFN response pathway

The first germline mutations impairing type I IFN immunity to be reported entail autosomal recessive complete STAT1 deficiency (Fig. 1 and table S1) (19). Cells from these patients do not signal via any STAT1-containing complex. Thus, they are refractory to stimulation by IL-27 and type II IFN (involving STAT1 homodimer formation). They also fail to respond to type I and III IFNs, where signaling is mediated by the ISGF-3 complex comprising STAT1, STAT2, and IRF9. Mutations in STAT2 and IRF9 reported later leave responses to IL-27 and type II IFN unimpaired (20, 21). The viral infections seen in patients with ISGF-3 complex deficiency were attributed to a specific defect in cellular responses to type I IFNs when they were observed in patients with mutations in IFNAR1 or IFNAR2 (encoding the type I IFN receptor chains) (22–26) but not IL-10RB (encoding a chain common to the receptors for IL-10, IL-22, IL-26, and type III IFNs) (27).

These viral infections are life-threatening but, unexpectedly, not as diverse as was predicted on the basis of the ubiquitous expression and pleiotropic function of type I IFNs. Particular susceptibilities relate to herpes simplex virus 1 encephalitis (HSE), critical influenza, COVID-19 pneumonia, and adverse reactions to live-attenuated

viral vaccines against measles and yellow fever. Remarkably, these patients do not demonstrate an overt phenotype beyond viral disease susceptibility, consistent with type I IFNs functioning as antiviral molecules rather than cytokines. Moreover, these individuals can apparently fend off a wide range of common RNA and DNA viruses, particularly viruses with a tropism outside of the respiratory tract and central nervous system (CNS). With fewer than 100 patients reported to date, most of whom live in developed countries, the breadth of infectious susceptibility of these patients is not yet fully determined. However, loss-of-function (LOF) alleles in IFNAR1 and IFNAR2—although very rare in most regions—are common in the Western Pacific and Arctic, respectively (28, 29). This suggests that human type I IFNs in the context of the whole organism living within a natural ecosystem (in natura) are redundant against many more viruses than anticipated from in vitro cell experimentation and in vivo mouse studies.

Deficits of individual type I IFNs and target genes

The mechanism by which these inborn errors in type I IFNs predispose individuals to infection is being progressively clarified by the identification of defects in genes that control the production of, or the response to, type I IFN subtypes, as well as by the description of autoantibodies (autoAbs) that neutralize specific type I IFNs (30, 31). Notably, autoAbs neutralizing IFN- ω or the 12 variants of IFN- α are sufficient to result in critical COVID-19, Middle East respiratory syndrome (MERS), or influenza pneumonia (32–36). They also underlie West Nile virus encephalitis (suggesting that now-undescribed inborn errors in type I IFNs may also predispose individuals to this condition) (37) and adverse reactions to yellow fever (38). There is no evidence, however, that such antibodies confer a susceptibility to measles vaccine (in contrast with patients with ISGF-3 complex deficiency). Indeed, measles vaccine is apparently not harmful even in children with autoAbs to IFN- α , IFN- β , and IFN- ω because of mutations in autoimmune regulator (AIRE) or other related inborn errors in central T cell tolerance (39, 40).

These observations suggest that the neutralization of these IFNs is not complete and/or ubiquitous (given that the expression of IFN- κ and IFN- ϵ is restricted to the skin and reproductive tract, respectively). No molecular defect in the 17 individual type I IFN genes has yet been reported. However, an inherited deficiency in IRF7 abolishes the induction of type I IFNs other than IFN- β (24, 41, 42). These rare patients apparently suffer only from respiratory viral infections, suggesting that IFN- β is protective against other viral infections, including in the CNS. Unexpectedly, mutations in IRF1, a transcription factor induced by STAT1-containing complexes, do not appear to predispose to common viral infections, with this deficit leading to an impairment of type II but not type I IFN

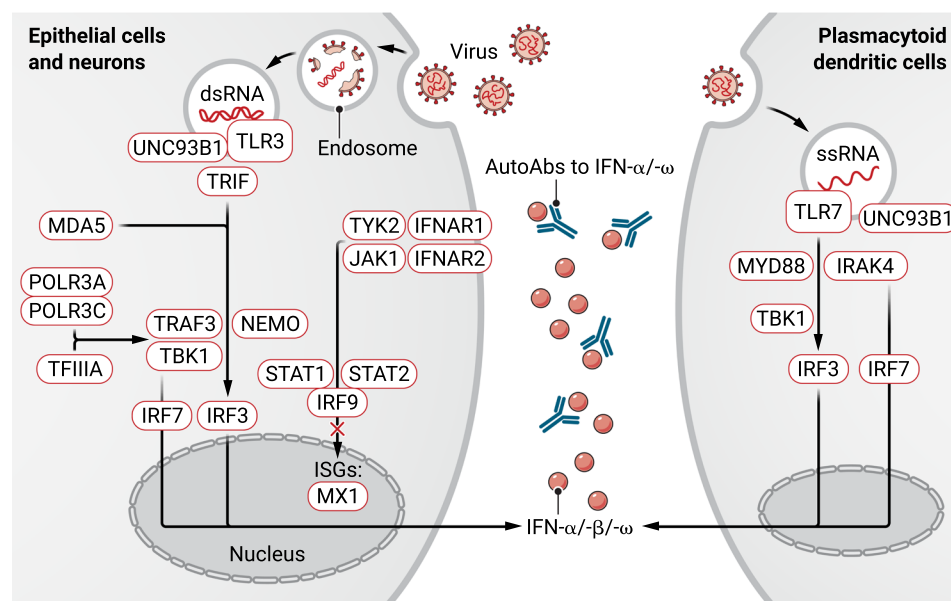


Fig. 1. Monogenic disorders underlying a deficit of type I IFN activity. The products of genes that are mutated in the germ line of patients with inborn errors resulting in type I IFN deficiency are shown. These proteins can be expressed in cells in which the virus can replicate (e.g., cortical neurons or respiratory epithelial cells) or in cells in which viral infection is abortive, yet the cells produce massive amounts of type I IFNs (e.g., pDCs). These proteins can impair the production of or the response to type I IFNs. AutoAbs neutralizing type I IFNs are also shown. These genetic disorders and their autoimmune phenocopies can underlie severe viral infections, including adverse reactions to live-attenuated viral vaccines and respiratory or cerebral viral infections.

activity (5). Notably, four (mutations in *ISG15*, *USP18*, *OAS1*, and *OAS2*) of the six known inborn errors in type I IFN-stimulated genes (ISGs) underlie inflammatory states and not viral disease susceptibility (43–45). Distinct mutations in *STAT2* lead to either an infectious susceptibility to virus (recessive deficiency in *STAT2* expression, hence deficient response to type I IFN via ISGF-3 formation) or an autoinflammatory state (recessive expression of mutant *STAT2* that has impaired the ability to facilitate the interaction of IFNAR2 with *USP18*, thereby enhancing responses to type I IFN via ISGF-3) (46, 47). Furthermore, heterozygous variants in interferon-induced GTP-binding protein *Mx1* (*MX1*) were found to underlie severe swine influenza cases in China (48). It is unclear whether mutations in *MX1* also predispose individuals to severe seasonal influenza or other respiratory viral diseases. The identification of mutations in the several hundred ISGs is expected to further clarify the mechanisms by which type I IFN deficits result in susceptibility to viral diseases (table S1) [and its expanded form (49)].

Deficits in type I IFN-inducing pathways

Much more is understood about inborn errors in genes that govern the induction of type I IFNs (i.e., as part of the viral recognition signaling machinery). Mutations in the Toll-like receptor 3 (TLR3) pathway (Fig. 1) can underlie HSE, critical influenza, or COVID-19 pneumonia (24, 50, 51). TLR3 is a double-stranded RNA (dsRNA) sensor localized in the endosomes of many cell types but not pDCs, the most potent producers of type I IFNs. Although TLR3 may sense viral intermediates and by-products, it also controls baseline type I IFN activity in certain cell types, including cortical neurons and respiratory epithelial cells, perhaps because of stimulation by endogenous agonists (52). To date, LOF mutations in the TLR7 pathway (Fig. 1) appear to selectively underlie critical COVID-19 pneumonia (53). TLR7 is an endosomal sensor of single-stranded RNA, and mutations in TLR7 profoundly impair pDC type I IFN production in response to severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). LOF genotypes in other nucleic acid TLR sensors, TLR8 and TLR9, have not yet been reported. However, patients with IL-1 receptor-associated kinase 4 or myeloid differentiation primary response protein (MyD88) deficiency—whose cells are unresponsive to TLR7, TLR8, and TLR9 agonists—are also prone to critical COVID-19 but, unexpectedly, not to other viral infections (54, 55). Rare patients with uncoordinated 93 homolog B1 (*UNC93B1*) deficiency, whose cells are unresponsive to TLR3, TLR7, TLR8, and TLR9, have so far only been reported to suffer from HSE (56).

Cytosolic sensors of viral dsRNA include melanoma differentiation-associated protein 5 (MDA5) and retinoic acid-inducible gene 1 protein (RIG-I). Although inherited MDA5 deficiency can underlie recurrent viral infections of the upper and lower respiratory tract (57, 58), as well as enterovirus encephalitis (59), no LOF mutations have been reported in RIG-I. Recessive deficiency of the cytosolic dsRNA sensors 2'-5'-oligoadenylate synthase 1 (*OAS1*) or *OAS2* or their downstream effector ribonuclease L (*RNaseL*) underlies SARS-CoV-2-triggered multisystem inflammatory syndrome (MIS-C), not because of excessive viral replication but because of excessive phagocytic inflammation (44). LOF mutations in the cyclic GMP-AMP synthase–stimulator of interferon genes (*cGAS-STING*) axis, the major cytosolic viral DNA sensing pathway, have not yet been described. By contrast, mutations in genes encoding *POLIII*, another DNA sensor, can predispose to varicella zoster virus encephalitis (60). Last, mutations in downstream *TBK1* (TANK-binding

kinase 1), *IRF3*, and *IRF7* can underlie severe viral diseases, albeit by disrupting multiple pathways inducing both type I and type III IFNs (41, 42, 61, 62).

Overall, inborn errors in the production of, or responses to, type I IFNs, as well as their autoimmune phenocopies, underlie an unexpectedly narrow range of life-threatening viral diseases. This observation probably reflects redundancy in antiviral effector mechanisms, both IFN (particularly type III) and non-IFN mediated. Those viral illnesses seen in the context of a failure of type I IFN immunity are acute and appear to be restricted to the respiratory tract and CNS except for adverse reactions to live-attenuated viral vaccines. A wide range of common viral diseases, particularly of the skin and digestive tract, appear to be normally controlled despite reduced or absent levels of biologically active type I IFNs.

INHERITED TYPE I INTERFERONOPATHIES

Definition of type I interferonopathies

The toxicity of type I IFNs was initially suggested by early studies in animal models (63). These data were strengthened by the description of increased levels of IFN- α activity in the sera and cerebrospinal fluid (CSF) of patients with Aicardi-Goutières syndrome (AGS) (64, 65) and reports of neurological disease and autoimmunity after iatrogenic exposure to IFN- α (66–69). The neurotoxic potential of type I IFNs was further emphasized by the observation of neuropathological features recapitulating those of AGS in transgenic mice chronically producing IFN- α from astrocytes (70). Definition of the genetic basis of AGS, implicating proteins involved in the metabolism (71–73), modification (74), and sensing (75, 76) of nucleic acids, led to the introduction of the type I interferonopathy concept, with these conditions defined as a set of monogenic diseases demonstrating enhanced type I IFN signaling, which directly contributes to the clinical phenotype (as opposed to representing a simple biomarker or a consequence of disease) (77). Points taken into account in assigning a mutant genotype as a type I interferonopathy are based on clinical and experimental data (Box 1). It should be stressed that the categorization of a disorder as a type I interferonopathy does not exclude its inclusion within another disease category. For example, diseases due to mutations in ataxia-telangiectasia mutated (*ATM*), *Artemis*, or Bloom syndrome *RecQ*-like helicase (*BLM*) are also appropriately considered disorders of DNA repair. An absence of patients described with inherited gain-of-function (GOF) variants in any of the 17 type I IFN genes or convincingly of either *IFNAR1* or *IFNAR2* precludes the categorical assertion that human type I IFNs (and which ones) are primary drivers of conditions now ascribed as type I interferonopathies. That said, recent data have linked the triplication of *IFNAR1/2* to a type I interferonopathic state in the context of trisomy 21 (78–82).

Categories of type I interferonopathies

We contend that there are now 52 known discrete mutant genotypes that can be defined as type I interferonopathies (Fig. 2 and table S2) [and its expanded form (49)], an increase from seven such disorders in 2011 (77). There are an additional 22 monogenic diseases where a link to enhanced type I IFN signaling has been suggested and/or is considered biologically plausible but which, in our view, is not yet sufficiently documented. We acknowledge that our categorization is open to debate and may change over time. The largest grouping of

Box 1. Points considered in assessing a mutant genotype as a type I interferonopathy.

1. Documentation of persistent up-regulation of ISG*, and/or IFN- α / β protein, expression in patient material (blood, PBMCs, CSF).
 2. Detailed in vitro and/or ex vivo cellular data supporting a biologically plausible mechanism to explain engagement of the type I IFN sensing or response machinery.
 3. Mutations reported in more than one gene involved in a shared biological pathway/function (e.g., alternative complement pathway, proteasome function, etc.).
 4. Mouse-cross data exploring a role for type I IFN sensing or response in disease in vivo.
 5. Association with the presence of clinical signs common to a number of type I interferonopathies, in particular, intracranial calcification, spasticity, chilblain-like vasculitic skin lesions, interstitial lung disease, Jaccoud arthropathy-like, and lupus-like disease.
 6. Evidence suggestive of a clinical response to the use of a JAK1 inhibitor or IFNAR blockade.
- PBMCs, peripheral blood mononuclear cells.

*In clinical practice, enhanced type I IFN signaling is most frequently assessed according to the expression of a subset of ISGs in blood leukocytes (184, 185). However, it should be noted that, despite differences in the expression of their respective receptors, most genes induced or regulated by type I IFNs are also controlled by type III IFNs via the same ISGF-3. Moreover, all such subsets express a type II IFN receptor, which signals via gamma-interferon activation factor (GAF) and not ISGF-3, resulting in the expression of an ill-defined overlapping set of target genes (186, 187).

type I interferonopathies relates to a disturbance in the regulation of ligands (most commonly nucleic acids), whereby abnormal quantity, modification, location, or sensing of ligand results in inappropriate IFN induction. A second important subgroup is accounted for by mutations in genes involved in regulating IFN signaling per se. Several comprehensive reviews exploring the pathways linking mutant genotype and enhanced type I IFN signaling are available (83–87). Here, we wish to draw attention to the recent observation of cross-talk between disorders of nuclear factor κ B (NF- κ B) and type I IFN signaling. Thus, although type I interferonopathies have been considered as a distinct subset within the broader classification of autoinflammatory diseases (88, 89), it is becoming evident that the situation is less clear-cut.

Notably, several disorders of ubiquitylation (90), associated with a loss of negative regulation of NF- κ B, demonstrate enhanced type I IFN signaling. IRF3—a critical transcription factor involved in type I IFN-mediated immune signaling—is negatively regulated by A20 through interaction with the NF- κ B-activating kinase TBK1 (91), possibly contributing to the observation of enhanced type I IFN signaling in the context of A20 haploinsufficiency (92–94). Furthermore, mutations in HOIL-1-interacting protein (HOIP)—the catalytic subunit of LUBAC (linear ubiquitination chain assembly complex)—and in OTULIN—a deubiquitinase that cleaves the Met1 polyubiquitination chains generated by LUBAC—are also associated with enhanced type I IFN signaling (95, 96). At least in the case of OTULIN deficiency, these findings may be explained by the demonstration that proteasome subunits are substrates of OTULIN deubiquitinase activity, with proteasome dysregulation in OTULIN-deficient cells leading to type I IFN activation (96, 97). In agreement

with these observations, dominant-negative mutations in the NF- κ B complex p65 subunit RELA have also recently been described in association with features of inflammation. These mutations drive the increased expression of *TLR7* and *MYD88* mRNA in pDCs and non-pDC myeloid cells leading to enhanced TLR7-mediated type I/III IFN signaling (98). These observations should probably not come as a surprise, given previous evidence of cross-talk between these signaling pathways (99–101).

Type I interferonopathies—Beyond type I IFN?

The toxic effect of leukocyte IFN- α in humans (66–69) and mice (63, 70), evidence from certain mouse crosses (102–104), the observation of a common set of clinical features (105), the narrow pattern of expression of type III IFNs and their receptor, and a lack of clinical overlap with phenotypes linked to excessive type II IFN activity support a primary role for type I IFNs in the pathogenesis of type I interferonopathies. However, as already noted, the lack of patients with GOF mutations in type I IFN genes per se or, definitively, in their receptor IFNAR warrants caution. Recognition of non-IFN-mediated signaling downstream of STING activation indicates that IFNs may not be the only drivers of nucleic acid-induced inflammation (106–108). Indeed, certain knockin mouse models of STING GOF (109, 110), but not all (111), suggest that type I IFN is irrelevant to the phenotype in this context. That said, differences between STING signaling in mice and humans make the importance of these observations uncertain (112–116).

Type I IFN signaling may contribute to a clinical phenotype, but not necessarily exclusively so, with disease due to LOF genotypes in DNASE2 particularly instructive here. Thus, mice null for deoxyribonuclease 2 accumulate undigested DNA in lysosomal macrophages, leading to chronic type I IFN signaling and lethal perinatal anemia (117). Although deletion of IFNAR1 rescued this phenotype (118), these double-knockout mice subsequently developed chronic arthritis (119). Both the embryonic and postnatal phenotypes are cGAS and STING dependent, indicating that dual type I IFN- and non-type I IFN-mediated pathways drive disease downstream of STING (120). Moreover, a single protein may have more than one biological function. For example, ADAR has both MDA5-mitochondrial antiviral signaling protein (MAVS)-dependent and IFIH1-MAVS-independent activity (121), whereas a failure of ribonucleotide excision repair because of loss of the RNase H2 complex leads to embryonic lethality in the absence of enhanced IFN signaling (122). The effect of anti-type I IFN treatments is expected to depend on the (relative) contribution of up-regulated type I IFN signaling to the associated clinical phenotype. Other factors (particularly the timing of treatment vis à vis disease onset and the accrual of nonreversible tissue damage, drug bioavailability, etc.) would also be expected to influence therapeutic efficacy.

**CROSS-TALK BETWEEN THE “UPS AND DOWNS”
Reciprocal genotypes**

One would predict the existence of “reciprocal genotypes” where distinct mutations in the same gene lead to a differential effect (down- or up-regulation) on type I IFN signaling. Indeed, on the basis of the list of monogenic disorders we have compiled (tables S1 and S2), five such examples have been described to date (Table 1). We do not include mutations in TBK1 (dominant deficiency predisposing to viral disease and recessive deficiency inflammation) (61,

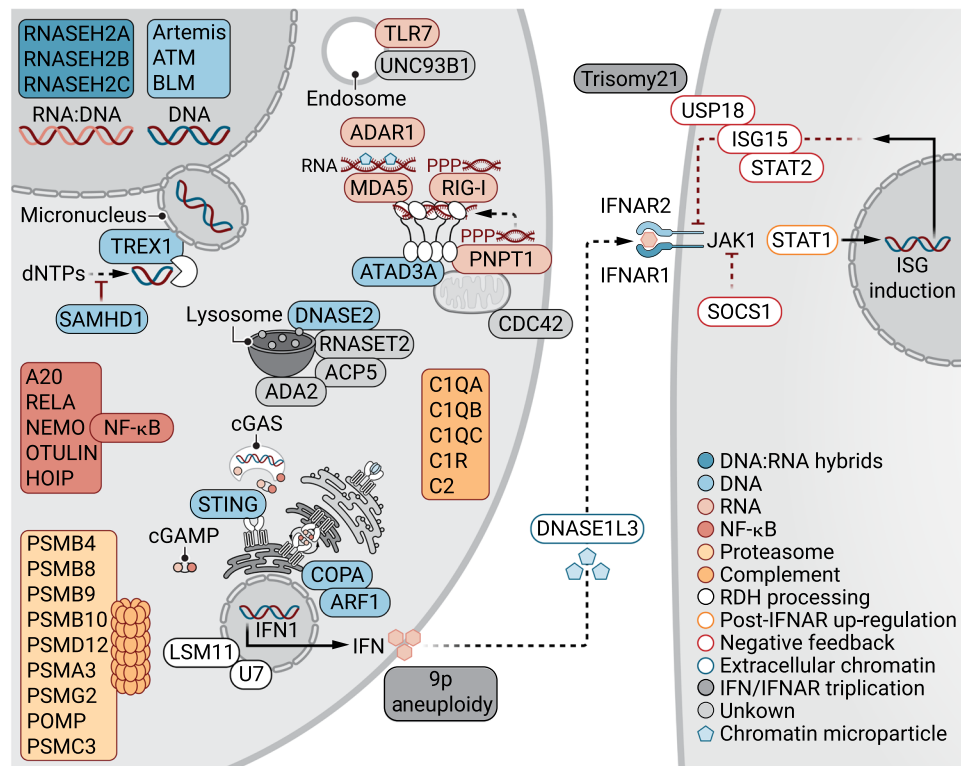


Fig. 2. Monogenic disorders underlying type I interferonopathies. The products of genes, which, when mutated, are associated with putative type I interferonopathies, are shown in the colored lozenges. In addition, trisomy 21 and 9p aneuploidy are associated with enhanced type I IFN signaling. The products of these genes are involved in the generation, metabolism, or sensing of nucleic acids (left) or in the regulation of the response to IFN after engagement of its receptor (right). Genes are grouped broadly according to function and/or organelle. dNTPs, deoxynucleotide triphosphates.

123) and OAS1/OAS2 (dominant GOF predisposing to inflammation and recessive deficiency MIS-C) (44, 124), despite their interest, because type I IFN only seems to be central to the pathogenesis of HSE in patients with dominant deficiency of TBK1. Reciprocal mutations in TLR7 and MDA5 are relatively straightforward to understand at a molecular level, with down- or up-regulation of type I IFN signaling explained by LOF or GOF, respectively. X-linked recessive TLR7 deficiency impairs pDC sensing of SARS-CoV-2 and underlies critical COVID-19 pneumonia in about 1% of male cases (53). By contrast, patients with TLR7 GOF exhibit features of systemic lupus erythematosus (SLE) with substantial neurological involvement (125, 126). Similarly, although autosomal recessive LOF of UNC93B1 predisposes individuals to HSE (56), GOF mutations result in SLE (127, 128), probably involving a disturbance of a delicate equilibrium in the interaction of UNC93B1 with TLR7, TLR8, TLR9, and other proteins (129–131). Autosomal recessive MDA5 deficiency can underlie recurrent respiratory viral and enteroviral disease of the CNS (57–59). By contrast, heterozygosity for GOF mutations in MDA5 can result in a spectrum of neurological and extraneurological features spanning AGS and Singleton-Merten syndrome (76, 132). In the case of STAT2, down- or up-regulation of IFN signaling both involve LOF at the molecular level, relating to distinct functions of the protein in either mediating IFN signaling or limiting IFN signaling through negative feedback (133). The situation with respect to STAT1 is more complex, likely reflecting the

broad cytokine disturbance consequent on loss or gain of activity (134). These disorders neatly illustrate that type I IFN activity governed by a single locus must be kept within a narrow window to prevent life-threatening viral or inflammatory conditions.

Unexpected inflammation and infection

Paradoxically, patients with type I IFN deficiency may display inflammatory episodes, which can probably be explained mechanistically in many cases. Thus, type I IFN deficiency is characterized by viral replication and spread in the first hours and days of infection. As a result, in an attempt to control viral spread, leukocytes are attracted to infected tissues but at the cost of clinical and biological inflammation. An excellent example of this phenomenon is the acute respiratory distress syndrome seen in patients with critical COVID-19 pneumonia because of IFNAR1 or IRF7 deficiency or autoAbs to type I IFNs (135). Inflammatory episodes in the absence of viral detection in such patients may attest to delayed medical attention, when viral clearance is eventually achieved after an unusually robust inflammatory response (136, 137). Another example is recessive deficiency in TBK1, which underlies a tumor necrosis factor- α (TNF- α) autoinflammatory phenotype (123). However, some episodes of autoinflammation in these individuals may be triggered by viral infections, as suggested by a case of lethal COVID-19 pneumonia (138).

Importantly, patients with type I interferonopathies are not generally prone to severe infections. Some genotypes may predispose to infection because they are associated with multiple cytokine dysfunction—most particularly STAT1 GOF—or a disturbance in DNA repair (e.g., mutations in ATM, Artemis, and BLM). Other instances include associations between alternative complement pathway deficiency and *Neisseria* infection and between the ISG15 LOF genotype and MSMD. Both examples are explained by specific functions of the relevant protein beyond any effect on IFN signaling: The terminal components of complement form the membrane attack complex, which pokes holes into *Neisseria*, and ISG15 is secreted as a type II IFN-inducing cytokine. Although rare reports of *Pneumocystis jirovecii* infection in the context of mutations in *STING1*, *COPA*, and *IFIH1* may reflect underlying lung damage, the possibility of a primary infectious susceptibility because of a defect in T cell function in SAVI (STING-associated vasculopathy with onset in infancy) exists (139–142). The fact that most individuals with other type I interferonopathies do not demonstrate a susceptibility to infection is notable given data suggesting that chronic exposure to enhanced type I IFN signaling can negatively affect adaptive immune function (143–145). It remains unclear whether patients with type I interferonopathies are protected from at least some viral

Table 1. Genotypes associated with reciprocal up and down type I IFN signaling. AI, autoimmunity; HLH, hemophagocytic lymphohistiocytosis; MM, molecular mechanism.

Gene	Down-regulation	MM	Up-regulation	MM
<i>IFIH1</i>	Viral respiratory disease; enterovirus rhombencephalitis	LOF	Interferonopathy (AGS)	GOF
<i>TLR7</i>	SARS-CoV-2 pneumonia	LOF	SLE	GOF
<i>UNC93B1</i>	HSE	LOF	SLE	Unclear
<i>STAT2</i>	HSE; influenza and SARS-CoV-2 pneumonia; live-attenuated viral vaccine strain infection; HLH-like	LOF	Interferonopathy (AGS-like)	LOF
<i>STAT1</i>	HSE; influenza and SARS-CoV-2 pneumonia; live-attenuated viral vaccine strain; MSMD	LOF	AI; mucocutaneous candidiasis; others	GOF

diseases, as would be predicted from their higher basal levels of type I IFNs. A disturbance in the dynamics of type I IFN regulation may also explain a lower incidence of viral disease, but increased infection-related morbidity and mortality, in trisomy 21 (80).

Quantitative biology of type I IFNs in the CNS

The study of inherited type I IFN deficits and type I interferonopathies indicates that too little or too much type I IFN is pathogenic. Perhaps the importance of type I IFN dosage is nowhere better illustrated than in the CNS. Although CNS-intrinsic and cortical neuron-intrinsic type I IFN is essential for protection against herpes simplex virus-1 (TLR3 pathway), enterovirus (MDA5), and West Nile virus (autoAbs to type I IFNs) encephalitis (50, 56, 60–62, 146, 147), excessive type I IFN production has neurotoxic potential, as inferred by the brain involvement and enhanced levels of CSF IFN- α observed in AGS (64) and the neurological sequelae of IFN- α therapy (66, 69, 148–150). In other words, the production of antiviral type I IFN by CNS-resident cells per se carries with it the risk of tissue damage, a point which may be particularly relevant for postmitotic neurons that have limited capacity for regeneration. Data exist to suggest a role for constitutive basal levels of IFN in CNS antiviral signaling (52, 151), possibly determined by a balance between background expression of endogenous ligands and the sensitivity (threshold) of their cognate receptors [illustrated, for example, by LOF or GOF mutations in ADAR and MDA5, respectively (74, 76)]. Indeed, the role of constitutive immune mechanisms [restriction factors, autophagy, and proteasomal degradation (152)] in controlling CNS infection—including those that when genetically deficient underlie viral encephalitis (e.g., mutations in *SNORA31*, *DBR1*, and *TMEFF1* (153–155))—may reflect the associated deleterious effects of type I IFN in the brain. More generally, the unexpectedly narrow range of viral infections in patients with type I IFN deficiency suggests that multiple type I IFN-independent mechanisms of intrinsic immunity are at work to fend off these and other viruses. These responses may have been evolutionarily driven by the risk of excessive type I IFNs, as attested by the growing number of type I interferonopathies associated with neurological involvement. Similar mechanisms may be relevant in other tissues.

OUTSTANDING QUESTIONS
Type I IFN: Quantity, duration, timing, and place

Type I IFN signaling can be considered in two states: constitutive (tonic) and stimulation induced. Both of these states can be disturbed, in an up or down fashion, in monogenic disease. Here, type I IFN-mediated immunological homeostasis reflects a cell type-specific balance of pro- and anti-inflammatory signals operating within a complex network of feedforward and feedback loops (80, 156–158). One notable aspect of this balance is the phenomenon of priming (i.e., the requirement for sequential stimuli), which involves type I IFN species per se and components of the type I IFN signaling machinery (particularly transcriptional factors). Another aspect is the fact that nucleic acid receptors are themselves inducible by type I IFNs. The quantity (amount), duration (i.e., acute versus chronic), timing (both in terms of sequenced events and organismal developmental stage), and place (i.e., cell, tissue, organ, etc.) of signaling are all relevant to global homeostasis (159–162). The tissue- and organ-specific nature of both type I IFN deficiencies and type I interferonopathies remains unresolved. The questions remain why viral infections of the brain and lungs predominate in type I IFN deficiencies without any detectable skin infections and why type I interferonopathies are broadly characterized as disorders of the brain, skin, and lung.

Tissue specificity is presumably determined by distinct expression patterns of disease-associated proteins, their protein partners, type I IFN-inducing signaling components, and proteins involved in alternative pro- and anti-inflammatory signaling pathways. Furthermore, clinical manifestation of disorders of reduced type I IFN signaling usually requires an environmental exposure (to vaccination or an infectious agent). This is not so obviously the case for type I interferonopathies, but such exposures, against the background of an already primed immune system, may also serve as disease triggers [and their absence may explain the clinical nonpenetrance observed in certain mutant genotypes (132, 163–165)]. Although IFN- α is present in the CSF and serum of a number of type I interferonopathies, the primary sites of IFN- α production in AGS (a neurologically focused phenotype) and SAVI (where neurological disease is rare) are the CNS and the periphery, respectively (166). STING signaling is central to seven of nine AGS-associated genotypes and all

SAVI cases. However, an important distinction is that endogenous nucleic acid ligands activate cGAS–2'3'-cyclic GMP-AMP (cGAMP)–STING signaling in AGS, whereas mutant STING proteins are constitutively active without a requirement for ligand engagement in SAVI.

Last, the question of what actually drives tissue damage in type I interferonopathies remains uncertain. Although the repertoire of genes induced by type I IFNs is likely key, some ISGs may be more important than others [e.g., Z-DNA binding protein 1 (ZBP1) in the context of ADAR1 deficiency (167)]. However, the observation of clinically nonpenetrant individuals carrying bona fide mutations and demonstrating persistently elevated ISG expression in blood is notable (132). These individuals are reminiscent of a mouse model where transgenic expression of a picornavirus RNA-dependent RNA polymerase leads to a marked up-regulation of ISG expression and profound viral resistance via an MDA5-MAVS-dependent pathway but are entirely healthy (168).

Additional phenotypes of type I IFN inborn errors

Given that many of the disorders discussed here have been identified only relatively recently, the evolution of these phenotypes over time remains to be defined. Inherited type I IFN deficiencies underlie viral diseases, often with associated inflammation and in the absence of detectable leukocytic dysfunction. Still, it is possible that other clinical phenotypes will be discovered in the future, particularly as patients are followed over time, further patients are diagnosed in diverse regions of the world, new deficits are identified, and novel viral infections are discovered. For example, certain type I IFN deficits may predispose some individuals to malignancy and/or alter the response to subsequent therapies. Indeed, type I IFN was the first cytokine to be used therapeutically in humans, with spectacular efficacy against particular tumors (e.g., hairy cell leukemia and Kaposi sarcoma). Moreover, it is intriguing that some type I interferonopathy genotypes seemingly underlie a relatively “clinically pure” autoinflammatory phenotype (e.g., proteasome-related syndromes, STING GOF, and ISG15 LOF), whereas others are consistently associated with autoimmune disease [e.g., LOF mutations in *ACP5* causing SLE (169)]. Still other genotypes demonstrate features of both phenotypes [e.g., COPI coat complex subunit alpha (COPA) syndrome and three prime repair exonuclease 1 (TREX1) LOF]. The distinction between autoinflammation versus autoimmunity among type I interferonopathies may be because of the impact of elevated type I IFN activity in leukocytes in the latter but not in the former. Indeed, other inborn errors of macrophages and adaptive lymphocytes have been implicated in the pathogenesis of autoimmune phenotypes including SLE (170). At any rate, the already broad phenotypic range of type I interferonopathies suggests that there may be hitherto unsuspected clinical manifestations to be discovered in both known and unreported type I interferonopathies. Last, although certain type I interferonopathies can predispose individuals to infection, it is unclear whether other conditions, such as allergy or malignancy, are more or less common in these patients.

Physiological range in other immune signaling pathways

Our understanding of human inherited type I IFN deficiencies and type I interferonopathies is unique, because there is no other cytokine or related immune molecule (e.g., type II IFN, growth factors, etc.) for which the physiological range has been genetically

delineated in natural conditions. This contrasts with the field of endocrinology, where conditions associated with decreased or elevated levels have been found for multiple hormones. Although inherited deficits have been described for many cytokines (e.g., type II IFN, IL-17A/F, etc.) together with their associated stereotyped susceptibilities to infection, inherited and selective GOF mutations of the corresponding activity have not yet been reported. Moreover, although selective GOF mutations of other cytokines have been identified (e.g., GOF in IL-1 β or TNF activity because of many autoinflammatory conditions that clinically respond to IL-1 or TNF blockade) with their corresponding inflammatory phenotype and response to treatment, selective genetic deficiencies of the same cytokines have yet to be discovered. For example, GOF mutations in NLRP3 underlie various autoinflammatory conditions, driven by excessive IL-1 β production, but humans with NLRP3 and IL1R deficiency have never been described (171). The search for inherited conditions defining the “ups” and “downs” of other cytokines would be expected to shed new light on the physiological ranges defining evolutionarily sculpted homeostasis. These results may be as unexpected as the narrow spectrum of life-threatening viral diseases in the type I IFN deficiencies and the spectrum of autoinflammation and autoimmunity related to type I interferonopathies. Such observations have forced a profound revision of the type I IFN paradigm established from the early 1960s onward. Similar conceptual shifts may occur when the ups and downs of “classic” molecules such as IL-1 β , TNF, and type II IFN are discovered. We predict that the field of immunology will benefit from these studies as much as endocrinology did when genetic studies of the ups and downs were determined to delineate the physiological and pathological impacts of various hormones.

MEDICINE OF TYPE I IFNS

The description of type I IFN deficiencies and type I interferonopathies suggests that type I IFN activity can be manipulated to within a physiological range to prevent or treat viral and inflammatory disease. With respect to viral diseases, type I IFN therapy, including IFN- α or IFN- β , in patients whose response pathway is intact (and taking into account any presence of specific IFN subtype autoAbs), is logical. Inhaled IFN- β is a particularly attractive approach to prevent or treat respiratory viral infections in type I IFN-deficient patients, because it would avoid systemic side effects and may protect patients with autoAbs to other IFNs (172). Type III IFNs may provide an alternative strategy if the JAK-STAT pathway is intact. Patients with type I IFN deficiencies should avoid viruses, be vaccinated against as many viruses as possible (excluding live viral vaccines), receive monthly immunoglobulin G infusions (which contain Abs to many common viruses), and be hospitalized in the case of a suspicion of infection, with prompt consideration of the possibility of life-threatening viral disease particularly of the lungs and brain. Given the nature of their infectious susceptibilities and the ubiquitous expression of the type I IFN receptor, the efficacy of hematopoietic stem cell transplant (HSCT) may be limited in these patients and disease specific. Nevertheless, HSCT has been tried in autosomal recessive STAT1 deficiency with encouraging results (173).

The treatment of type I interferonopathies has, relatively speaking, proved more straightforward. Before the definition of the underlying molecular defects, disease in patients with STING and

proteasomal dysfunction had proven refractory to a range of immunomodulatory agents, including IL-1 β , IL-6, and TNF- α blockers (174–176). By contrast, the use of JAK1/2 and JAK1/3 inhibitors—premised on the blockade of JAK1, which is constitutively associated with the type I IFN receptor and indispensable for its activity—has demonstrated a benefit in a number of discrete type I interferonopathies, indicating that these drugs address a relevant biological pathway (177, 178). Despite the lack of absolute specificity of these inhibitors and the involvement of JAKs (including JAK1 itself) in other cytokine receptor pathways, side effects have been relatively modest. That said, the effects have not been equivalent to those observed with anti-IL-1 β therapy in patients with NLRP3 mutations, for example (179). Relative treatment failure may relate to the stage of disease at which therapy is initiated and the accessibility of drugs to the disease-relevant tissue (180, 181). In this regard, the use of intrathecal treatments in AGS is potentially attractive (182). Experience with HSCT in type I interferonopathies remains extremely limited, with the neurological involvement observed in many of these diseases an additional concern when considering efficacy. The search for a precision medicine approach, including targeting a therapy according to the specific underlying molecular defect (e.g., blocking STING or MAVS in the case of aberrant cytosolic DNA or RNA signaling, respectively), remains an exciting goal.

CONCLUDING REMARKS

The study of human inborn errors underlying insufficient or excessive type I IFNs has been enlightening. There is a narrow physiological range compatible with, and essential for, life (183). It is remarkable that there is no other cytokine, and perhaps no other molecule involved in immunity, for which the ups and downs have been so well genetically defined in humans (i.e., in natura). Through the study of these diseases, we have learned that type I IFNs are not true cytokines in the narrow sense of the term. That is to say, there does not seem to be a detectable leukocytic phenotype, developmentally or functionally, in patients with a deficit in type I IFN signaling. Type I IFNs operate as antivirals in cells other than leukocytes, although their production by pDCs matters greatly. These molecules operate in a cell-intrinsic (e.g., via IFN produced by cortical neurons or respiratory epithelial cells) and cell-extrinsic manner (e.g., via pDCs recruited to the brain or the lungs). It must be reiterated that there have been no viral infections of leukocytes in patients deprived of type I IFNs so far. Type I IFNs are also apparently redundant in many nonleukocytic cells, given that viral infections of the brain and lungs are the predominant phenotype. The considerable level of redundancy at the whole organism level is unexpected and, together with their deleterious impact, suggests that multiple type I IFN-independent mechanisms of immunity to viruses must exist. Who would have predicted that LOF alleles of IFNAR1 or IFNAR2 would reach a minor allele frequency of between 1 and 10% in certain populations, which, although isolated, harbor hundreds of human-tropic viruses? Moreover, not all type I IFNs were created equal. It has now been shown that IFN- ω , probably the least studied type I IFN, does matter physiologically. What do human IFN- ϵ and IFN- κ really do? Mutants of either of these type I IFNs would illuminate their function. The fundamental antiviral role of type I IFN also raises the question of why it originated in jawed vertebrates. Is the emergence of adaptive immunity at a similar evolutionary stage a coincidence? If so, are there IFN-like genes in jawless fish, driven by

convergent evolution? By contrast, we have learned that type I IFNs can be harmful, perhaps even more so than predicted by early work in animal models and the iatrogenic effects of type I IFN therapy in humans. Immune mechanisms that effectively combat infection are fundamental to survival. Type I interferonopathies illustrate the risk inherent in the evolutionary adoption of nucleic acid detection as a primary antiviral response. When seen in this light, the astounding diversity of associated autoinflammatory and autoimmune phenotypes present in this set of disorders is not wholly unanticipated.

Supplementary Materials

This PDF file includes:

Tables S1 and S2

REFERENCES AND NOTES

1. A. Isaacs, J. Lindenmann, Virus interference. I. The interferon. *Proc. R. Soc. Lond. B Biol. Sci.* **147**, 258–267 (1957).
2. A. Isaacs, J. Lindenmann, R. C. Valentine, Virus interference. II. Some properties of interferon. *Proc. R. Soc. Lond. B Biol. Sci.* **147**, 268–273 (1957).
3. B. Reizis, Plasmacytoid dendritic cells: Development, regulation, and function. *Immunity* **50**, 37–50 (2019).
4. C. F. Nathan, H. W. Murray, M. E. Wiebe, B. Y. Rubin, Identification of interferon-gamma as the lymphokine that activates human macrophage oxidative metabolism and antimicrobial activity. *J. Exp. Med.* **158**, 670–689 (1983).
5. J. Rosain, A. L. Neehus, J. Manry, R. Yang, J. Le Pen, W. Daher, Z. Liu, Y. H. Chan, N. Tahiul, O. Turel, M. Bourgey, M. Ogishi, J. M. Doisne, H. M. Izquierdo, T. Shirasaki, T. Le Voyer, A. Guerin, P. Bastard, M. Moncada-Velez, J. E. Han, T. Khan, F. Rapaport, S. H. Hong, A. Cheung, K. Haake, B. C. Mindt, L. Perez, Q. Philippot, D. Lee, P. Zhang, D. Rinchai, F. Al Ali, M. M. A. Ata, M. Rahman, J. N. Peel, S. Heissel, H. Molina, Y. Kendir-Demirkol, R. Bailey, S. Zhao, J. Bohlen, M. Mancini, Y. Seeleuthner, M. Roelens, L. Lorenzo, C. Soudee, M. E. J. Paz, M. L. Gonzalez, M. Jeljeli, J. Soulier, S. Romana, A. S. L. Honneur, M. Materna, R. Martinez-Barricarte, M. Pochon, C. Oleaga-Quintas, A. Michiev, M. Migaud, R. Levy, M. A. Alyanekian, F. Rozenberg, C. A. Croft, G. Vogt, J. F. Emile, L. Kremer, C. S. Ma, J. H. Fritz, S. M. Lemon, A. N. Spaan, N. Manel, L. Abel, M. R. MacDonald, S. Boisson-Dupuis, N. Marr, S. G. Tangye, J. P. Di Santo, Q. Zhang, S. Y. Zhang, C. M. Rice, V. Beziat, N. Lachmann, D. Langlais, J. L. Casanova, P. Gros, J. Bustamante, Human IRF1 governs macrophagic IFN-gamma immunity to mycobacteria. *Cell* **186**, 621–645.e33 (2023).
6. J. L. Casanova, J. D. MacMicking, C. F. Nathan, Interferon- γ and infectious diseases: Lessons and prospects. *Science* **384**, eadi1016 (2024).
7. I. Gresser, Wherefore interferon? *J. Leukoc. Biol.* **61**, 567–574 (1997).
8. H. M. Lazear, J. W. Schoggins, M. S. Diamond, Shared and distinct functions of type I and type III interferons. *Immunity* **50**, 907–923 (2019).
9. J. Vilcek, Interferon research BC (before cloning). *Curr. Top. Microbiol. Immunol.* **316**, 9–22 (2007).
10. I. Gresser, Interferon: An unfolding tale. *J. Interferon Cytokine Res.* **27**, 447–452 (2007).
11. J. L. Casanova, I. Gresser. *J. Interferon Cytokine Res.* **39**, 317–320 (2019).
12. H. H. Hoffmann, W. M. Schneider, C. M. Rice, Interferons and viruses: An evolutionary arms race of molecular interactions. *Trends Immunol.* **36**, 124–138 (2015).
13. D. W. LaFleur, B. Nardelli, T. Tsareva, D. Mather, P. Feng, M. Semenuk, K. Taylor, M. Buerger, D. Chinchilla, V. Roshke, G. Chen, S. M. Ruben, P. M. Pitha, T. A. Coleman, P. A. Moore, Interferon-kappa, a novel type I interferon expressed in human keratinocytes. *J. Biol. Chem.* **276**, 39765–39771 (2001).
14. Z. R. C. Marks, N. K. Campbell, N. E. Mangan, C. J. Vandenberg, L. J. Gearing, A. Y. Matthews, J. A. Gould, M. D. Tate, G. Wray-McCann, L. Ying, S. Rosli, N. Brockwell, B. S. Parker, S. S. Lim, M. Bilandzic, E. L. Christie, A. N. Stephens, E. de Geus, M. J. Wakefield, G. Y. Ho, O. McNally, Australian Ovarian Cancer Study, I. A. McNeish, D. D. L. Bowtell, N. A. de Weerd, C. L. Scott, N. M. Bourke, P. J. Hertzog, Interferon- ϵ is a tumour suppressor and restricts ovarian cancer. *Nature* **620**, 1063–1070 (2023).
15. P. Boudinot, C. Langevin, C. J. Secombes, J. P. Levraud, The peculiar characteristics of fish type I interferons. *Viruses* **8**, 298 (2016).
16. C. J. Secombes, J. Zou, Evolution of interferons and interferon receptors. *Front. Immunol.* **8**, 209 (2017).
17. O. D. Adeyemi, Y. Tian, C. N. Khwatenge, L. Grayfer, Y. Sang, Molecular diversity and functional implication of amphibian interferon complex: Remarkable immune adaptation in vertebrate evolution. *Dev. Comp. Immunol.* **140**, 104624 (2023).
18. Y. Sang, Q. Liu, J. Lee, W. Ma, D. S. McVey, F. Blecha, Expansion of amphibian intronless interferons revises the paradigm for interferon evolution and functional diversity. *Sci. Rep.* **6**, 29072 (2016).

19. S. Dupuis, E. Jouanguy, S. Al-Hajjar, C. Fieschi, I. Z. Al-Mohsen, S. Al-Jumaah, K. Yang, A. Chapgier, C. Eidenschenck, P. Eid, A. Al Ghonaim, H. Tufenkeji, H. Frayha, S. Al-Gazlan, H. Al-Rayes, R. D. Schreiber, I. Gresser, J. L. Casanova, Impaired response to interferon- α/β and lethal viral disease in human STAT1 deficiency. *Nat. Genet.* **33**, 388–391 (2003).
20. S. Hambleton, S. Goodbourn, D. F. Young, P. Dickinson, S. M. Mohamad, M. Valappil, N. McGovern, A. J. Cant, S. J. Hackett, P. Ghazal, N. V. Morgan, R. E. Randall, STAT2 deficiency and susceptibility to viral illness in humans. *Proc. Natl. Acad. Sci. U.S.A.* **110**, 3053–3058 (2013).
21. N. Hernandez, I. Melki, H. Jing, T. Habib, S. S. Y. Huang, J. Danielson, T. Kula, S. Drutman, S. Belkaya, V. Rattina, L. Lorenzo-Diaz, A. Boulai, Y. Rose, N. Kitabayashi, M. P. Rodero, C. Dumaine, S. Blanche, M. N. Lebras, M. C. Leung, L. S. Mathew, B. Boisson, S. Y. Zhang, S. Boisson-Dupuis, S. Giliani, D. Chaussabel, L. D. Notarangelo, S. J. Elledge, M. J. Ciancanelli, L. Abel, Q. Zhang, N. Marr, Y. J. Crow, H. C. Su, J. L. Casanova, Life-threatening influenza pneumonitis in a child with inherited IRF9 deficiency. *J. Exp. Med.* **215**, 2567–2585 (2018).
22. C. J. Duncan, S. M. Mohamad, D. F. Young, A. J. Skelton, T. R. Leahy, D. C. Munday, K. M. Butler, S. Morfopoulou, J. R. Brown, M. Hubank, J. Connell, P. J. Gavin, C. McMahon, E. Dempsey, N. E. Lynch, T. S. Jacques, M. Valappil, A. J. Cant, J. Breuer, K. R. Engelhardt, R. E. Randall, S. Hambleton, Human IFNAR2 deficiency: Lessons for antiviral immunity. *Sci. Transl. Med.* **7**, 307ra154 (2015).
23. N. Hernandez, G. Bucciol, L. Moens, J. Le Pen, M. Shahrooei, E. Goudouris, A. Shirkani, M. Changi-Ashtiani, H. Rokni-Zadeh, E. H. Sayar, I. Reisli, A. Lefevre-Utile, D. Zijlmans, A. Jurado, R. Pholien, S. Drutman, S. Belkaya, A. Cobat, R. Boudewijns, D. Jochmans, J. Neyts, Y. Seeleuthner, L. Lorenzo-Diaz, C. Enemchukwu, I. Tietjen, H. H. Hoffmann, M. Momenilandi, L. Poyhonen, M. M. Siqueira, S. M. B. de Lima, D. C. de Souza Matos, A. Homma, M. L. S. Maia, T. A. da Costa Barros, P. M. N. de Oliveira, E. C. Mesquita, R. Gijssbers, S. Y. Zhang, S. J. Seligman, L. Abel, P. Hertzog, N. Marr, R. M. Martins, I. Meyts, Q. Zhang, M. R. MacDonald, C. M. Rice, J. L. Casanova, E. Jouanguy, X. Bossuyt, Inherited IFNAR1 deficiency in otherwise healthy patients with adverse reaction to measles and yellow fever live vaccines. *J. Exp. Med.* **216**, 2057–2070 (2019).
24. Q. Zhang, P. Bastard, Z. Liu, J. Le Pen, M. Moncada-Velez, J. Chen, M. Ogishi, I. K. D. Sabli, S. Hodelib, C. Korol, J. Rosain, K. Bilguvar, J. Ye, A. Bolze, B. Bigio, R. Yang, A. A. Arias, Q. Zhou, Y. Zhang, F. Onodi, S. Korniotis, L. Karpf, Q. Philippot, M. Chbihi, L. Bonnet-Madin, K. Dorgham, M. Smith, W. M. Schneider, B. S. Razoaky, H. H. Hoffmann, E. Michailidis, L. Moens, J. E. Han, L. Lorenzo, L. Bizen, P. Meade, A. L. Neehus, A. C. Ugurbil, A. Corneau, G. Kerner, P. Zhang, F. Rapaport, Y. Seeleuthner, J. Manry, C. Masson, Y. Schmitt, A. Schluter, T. Le Voyer, T. Khan, J. Li, J. Fellay, L. Roussel, M. Shahrooei, M. F. Alosaimi, D. Mansouri, H. Al-Saud, F. Al-Mulla, F. Almourfi, S. Z. Al-Muhsen, F. Alshohime, S. Al Turki, R. Hasanato, D. van de Beek, A. Biondi, L. R. Bettini, M. D'Angio, P. Bonfanti, L. Imberti, A. Sottini, S. Paghera, E. Quiros-Roldan, C. Rossi, A. J. Oler, M. F. Tompkins, C. Alba, I. Vandermoot, J. C. Goffard, G. Smits, I. Migeotte, F. Haerynck, P. Soler-Palacin, A. Martin-Nalda, R. Colobran, P. E. Morange, S. Keles, F. Colkesen, T. Ozcelik, K. K. Yasar, S. Senoglu, S. N. Karabela, C. Rodriguez-Gallego, G. Novelli, S. Hraiech, Y. Tandjaoui-Lambiotte, X. Duval, C. Laouenan, C.-S. Clinicians, COVID-STORM Clinicians, COVID Clinicians, Imagine COVID Group, French COVID Cohort Study Group, CoV-Contact Cohort, Amsterdam UMC Covid-19 Biobank, COVID Human Genetic Effort, NIAID-USUHS/TAGC COVID Immunity, A. L. Snow, C. L. Dalgard, J. D. Milner, D. C. Vinh, T. H. Mogensen, N. Marr, A. N. Spaan, B. Boisson, S. Boisson-Dupuis, J. Bustamante, A. Puel, M. J. Ciancanelli, I. Meyts, T. Maniatis, Y. Soumelis, A. Amara, M. Nussenzweig, A. Garcia-Sastre, F. Krammer, A. Pujol, D. Duffy, R. P. Lifton, S. Y. Zhang, G. Gorochov, V. Beziat, E. Jouanguy, V. Sancho-Shimizu, C. M. Rice, L. Abel, L. D. Notarangelo, A. Cobat, H. C. Su, J. L. Casanova, Inborn errors of type I IFN immunity in patients with life-threatening COVID-19. *Science* **370**, eabd4570 (2020).
25. P. Bastard, J. Manry, J. Chen, J. Rosain, Y. Seeleuthner, O. AbuZaitun, L. Lorenzo, T. Khan, M. Hasek, N. Hernandez, B. Bigio, P. Zhang, R. Levy, S. Shrot, E. J. G. Reino, Y. S. Lee, S. Bouchet, M. Aubart, R. Gijssbers, V. Beziat, Z. Li, S. Pellegrini, F. Rozenberg, N. Marr, I. Meyts, B. Boisson, A. Cobat, J. Bustamante, Q. Zhang, E. Jouanguy, L. Abel, R. Somech, J. L. Casanova, S. Y. Zhang, Herpes simplex encephalitis in a patient with a distinctive form of inherited IFNAR1 deficiency. *J. Clin. Invest.* **131**, e139980 (2021).
26. Q. Zhang, D. Matuozzo, J. Le Pen, D. Lee, L. Moens, T. Asano, J. Bohlen, Z. Liu, M. Moncada-Velez, Y. Kendir-Demirkol, H. Jing, L. Bizen, A. Marchal, H. Abolhassani, S. Delafontaine, G. Bucciol, C. H. G. Effort, G. I. Bayhan, S. Keles, A. Kiykim, S. Hancerli, F. Haerynck, B. Florin, N. Hatipoglu, T. Ozcelik, G. Morelle, M. Zatz, L. F. P. Ng, D. C. Lye, B. E. Young, Y. S. Leo, C. L. Dalgard, R. P. Lifton, L. Renia, I. Meyts, E. Jouanguy, L. Hammarstrom, Q. Pan-Hammarstrom, B. Boisson, P. Bastard, H. C. Su, S. Boisson-Dupuis, L. Abel, C. M. Rice, S. Y. Zhang, A. Cobat, J. L. Casanova, Recessive inborn errors of type I IFN immunity in children with COVID-19 pneumonia. *J. Exp. Med.* **219**, (2022).
27. C. B. Korol, S. Belkaya, F. Alshohime, L. Lorenzo, S. Boisson-Dupuis, J. Brancalle, A. L. Neehus, S. Vilarinho, A. Zobaïda, R. Halwani, S. Al-Muhsen, J. L. Casanova, E. Jouanguy, Fulminant viral hepatitis in two siblings with inherited IL10RB deficiency. *J. Clin. Immunol.* **43**, 406–420 (2023).
28. P. Bastard, K. C. Hsiao, Q. Zhang, J. Choin, E. Best, J. Chen, A. Gervais, L. Bizen, M. Materna, C. Harmant, M. Roux, N. L. Hawley, D. E. Weeks, S. T. McGarvey, K. Sandoval, C. Barberena-Jonas, C. D. Quinto-Cortes, E. Hagelberg, A. J. Mentzer, K. Robson, B. Coulibaly, Y. Seeleuthner, B. Bigio, Z. Li, G. Uze, S. Pellegrini, L. Lorenzo, Z. Sbihi, S. Latour, M. Besnard, T. Adam de Beaumais, E. Jacqz-Aigrain, V. Beziat, R. Deka, L. Esera Tulifau, S. Viali, M. S. Reupena, T. Naseri, P. McNaughton, V. Sarkozy, J. Peake, A. Blincoe, S. Primhak, S. Stables, K. Gibson, S. T. Woon, K. M. Drake, A. V. S. Hill, C. Y. Chan, R. King, R. Ameratunga, I. Teiti, M. Aubry, V. M. Cao-Lormeau, S. G. Tangye, S. Y. Zhang, E. Jouanguy, P. Gray, L. Abel, A. Moreno-Estrada, R. L. Minster, L. Quintana-Murci, A. C. Wood, J. L. Casanova, A loss-of-function IFNAR1 allele in Polynesia underlies severe viral diseases in homozygotes. *J. Exp. Med.* **219**, e20220028 (2022).
29. C. J. A. Duncan, M. K. Skouboe, S. Howarth, A. K. Hollensen, R. Chen, M. L. Borresen, B. J. Thompson, J. Stremenova-Spegarova, C. F. Hatton, F. F. Staeger, M. K. Andersen, J. Whittaker, S. R. Paludan, S. E. Jorgensen, M. K. Thomsen, J. G. Mikkelsen, C. Heilmann, D. Buhas, N. F. Obro, J. T. Bay, H. V. Marquart, M. T. de la Morena, J. A. Klejka, M. Hirschfeld, L. Borgwardt, I. Forss, T. Masmas, A. Poulsen, F. Noya, G. Rouleau, T. Hansen, S. Zhou, A. Albrechtsen, R. Alizadehfar, E. J. Allenspach, S. Hambleton, T. H. Mogensen, Life-threatening viral disease in a novel form of autosomal recessive IFNAR2 deficiency in the Arctic. *J. Exp. Med.* **219**, e20212427 (2022).
30. K. E. Mogensen, P. Daubas, I. Gresser, D. Sereni, B. Varet, Patient with circulating antibodies to alpha-interferon. *Lancet* **2**, 1227–1228 (1981).
31. B. Pozzetto, K. E. Mogensen, M. G. Tovey, I. Gresser, Characteristics of autoantibodies to human interferon in a patient with varicella-zoster disease. *J. Infect. Dis.* **150**, 707–713 (1984).
32. P. Bastard, L. B. Rosen, Q. Zhang, E. Michailidis, H. H. Hoffmann, Y. Zhang, K. Dorgham, Q. Philippot, J. Rosain, V. Beziat, J. Manry, E. Shaw, L. Haljasmagi, P. Peterson, L. Lorenzo, L. Bizen, S. Trouillet-Assant, K. Dobbs, A. A. de Jesus, A. Belot, A. Kallaste, E. Catherineot, Y. Tandjaoui-Lambiotte, J. Le Pen, G. Kerner, B. Bigio, Y. Seeleuthner, R. Yang, A. Bolze, A. N. Spaan, O. M. Delmonte, M. S. Abers, A. Aiuti, G. Casari, V. Lampasona, L. Piemonti, F. Ciceri, K. Bilguvar, R. P. Lifton, M. Vasse, D. M. Smadja, M. Migaud, J. Hadjadj, B. Terrier, D. Duffy, L. Quintana-Murci, D. van de Beek, L. Roussel, D. C. Vinh, S. G. Tangye, F. Haerynck, D. Dalmau, J. Martinez-Picado, P. Brodin, M. C. Nussenzweig, S. Boisson-Dupuis, C. Rodriguez-Gallego, G. Vogt, T. H. Mogensen, A. J. Oler, J. Gu, P. D. Burbelo, J. I. Cohen, A. Biondi, L. R. Bettini, M. D'Angio, P. Bonfanti, P. Rossignol, J. Mayaux, F. Rieux-Laucat, E. S. Husebye, F. Fusco, M. V. Ursini, L. Imberti, A. Sottini, S. Paghera, E. Quiros-Roldan, C. Rossi, R. Castagnoli, D. Montagna, A. Licari, G. L. Marsaglia, X. Duval, J. Ghosn, HGID Lab, NIAID-USUHS Immune Response to COVID Group, COVID Clinicians, COVID-STORM Clinicians, Imagine COVID Group, French COVID Cohort Study Group, Milieu Interieur Consortium, CoV-Contact Cohort, Amsterdam UMC Covid-19 Biobank, COVID Human Genetic Effort, J. S. Tsang, R. Goldbach-Mansky, K. Kisand, M. S. Lionakis, A. Puel, S. Y. Zhang, S. M. Holland, G. Gorochov, E. Jouanguy, C. M. Rice, A. Cobat, L. D. Notarangelo, L. Abel, H. C. Su, J. L. Casanova, Autoantibodies against type I IFNs in patients with life-threatening COVID-19. *Science* **370**, eabd4585 (2020).
33. P. Bastard, A. Gervais, T. Le Voyer, J. Rosain, Q. Philippot, J. Manry, E. Michailidis, H. H. Hoffmann, S. Eto, M. Garcia-Prat, L. Bizen, A. Parra-Martinez, R. Yang, L. Haljasmagi, M. Migaud, K. Sarekannu, J. Maslovskaja, N. de Prost, Y. Tandjaoui-Lambiotte, C. E. Luyt, B. Amador-Borrero, A. Gaudet, J. Poissy, P. Morel, P. Richard, F. Cognasse, J. Troya, S. Trouillet-Assant, A. Belot, K. Saker, P. Garcon, J. G. Riviere, J. C. Lagier, S. Gentile, L. B. Rosen, E. Shaw, T. Morio, J. Tanaka, D. Dalmau, P. L. Tharaux, D. Sene, A. Stepanian, B. Megarbane, V. Triantafyllia, A. Fekkar, J. R. Heath, J. L. Franco, J. M. Anaya, J. Sole-Violan, L. Imberti, A. Biondi, P. Bonfanti, R. Castagnoli, O. M. Delmonte, Y. Zhang, A. L. Snow, S. M. Holland, C. Biggs, M. Moncada-Velez, A. A. Arias, L. Lorenzo, S. Bouchet, B. Coulibaly, D. Anglicheau, A. M. Planas, F. Haerynck, S. Duvlis, R. L. Nussbaum, T. Ozcelik, S. Keles, A. Bousfiha, J. El Bakkouri, C. Ramirez-Santana, S. Paul, Q. Pan-Hammarstrom, L. Hammarstrom, A. Dupont, A. Kurolop, C. N. Metz, A. Aiuti, G. Casari, V. Lampasona, F. Ciceri, L. A. Barreiros, E. Dominguez-Garrido, M. Vidigal, M. Zatz, D. van de Beek, S. Sahanic, I. Tancevski, Y. Stepanovskyy, O. Boyarchuk, Y. Nukui, M. Tsumura, L. Vidaur, S. G. Tangye, S. Burrell, D. Duffy, L. Quintana-Murci, A. Klocperk, N. Y. Kann, A. Shcherbina, Y. L. Lau, D. Leung, M. Coulangeat, J. Marlet, R. Koning, L. F. Reyes, A. Chauvineau-Grenier, F. Venet, G. Monneret, M. C. Nussenzweig, R. Arrestier, I. Boudhabhay, H. Baris-Feldman, D. Hagin, J. Wauters, I. Meyts, A. H. Dyer, S. P. Kennedy, N. M. Bourke, R. Halwani, N. S. Sharif-Askari, K. Dorgham, J. Sallet, S. M. Sedkaoui, S. Alkhater, R. Rigo-Bonnin, F. Morandeira, L. Roussel, D. C. Vinh, S. R. Ostrowski, A. Condino-Neto, C. Prando, A. Bonradenko, A. N. Spaan, L. Gilardin, J. Fellay, S. Lyonnet, K. Bilguvar, R. P. Lifton, S. Mane, HGID Lab, COVID Clinicians; COVID-STORM Clinicians, NIAID Immune Response to COVID Group, NH-COVID Study Group, Danish CHGE; Danish Blood Donor Study, St. James's Hospital; SARS CoV2 Interest group, French COVID Cohort Study Group; Imagine COVID-Group, Milieu Interieur Consortium; CoV-Contact Cohort; Amsterdam UMC Covid-19, Biobank Investigators, COVID Human Genetic Effort, CONSTANCES cohort, 3C-Dijon Study, Cerba Health-Care, Etablissement du Sang study group, M. S. Anderson, B. Boisson, V. Beziat, S. Y. Zhang, E. Vandreakos, O. Hermine, A. Pujol, P. Peterson,

- T. H. Mogensen, L. Rowen, J. Mond, S. Debette, X. de Lamballerie, X. Duval, F. Mentre, M. Zins, P. Soler-Palacin, R. Colobran, G. Gorochov, X. Solanich, S. Susen, J. Martinez-Picado, D. Raoult, M. Vasse, P. K. Gregersen, L. Piemonti, C. Rodriguez-Gallego, L. D. Notarangelo, H. C. Su, K. Kisand, S. Okada, A. Puel, E. Jouanguy, C. M. Rice, P. Tiberghien, Q. Zhang, A. Cobat, L. Abel, J. L. Casanova, Autoantibodies neutralizing type I IFNs are present in ~4% of uninfected individuals over 70 years old and account for ~20% of COVID-19 deaths. *Sci. Immunol.* **6**, eabl4340 (2021).
34. Q. Zhang, A. Pizzorno, L. Miorin, P. Bastard, A. Gervais, T. Le Voyer, L. Bizien, J. Manry, J. Rosain, Q. Philippot, K. Goavec, B. Padey, A. Cupic, E. Laurent, K. Saker, M. Vanke, K. Sarekannu, COVID Human Genetic Effort, Etablissement Français du Sang Study Group, Constances Cohort; 3C-Dijon Study, Cerba HealthCare Group, Lyon Antigrippe Working Group, REIPI INF Working Group, T. Garcia-Salum, M. Ferres, N. Le Corre, J. Sanchez-Céspedes, M. Balsera-Manzanero, J. Carratala, P. Retamar-Gentil, G. Abelda-Alonso, A. Valiente, P. Tiberghien, M. Zins, S. Debette, I. Meyts, F. Haerynck, R. Castagnoli, L. D. Notarangelo, L. I. Gonzalez-Granado, N. Dominguez-Pinilla, E. Andreaskos, V. Triantafyllia, C. Rodriguez-Gallego, J. Sole-Violan, J. J. Ruiz-Hernandez, F. R. de Castro, J. Ferreres, M. Briones, J. Wauters, L. Vanderbeke, S. Feys, C. Y. Kuo, W. T. Lei, C. L. Ku, G. Tal, A. Etzioni, S. Hanna, T. Fournet, J. S. Casalegno, G. Queromes, L. Argaud, E. Javouhey, M. Rosa-Calatrava, E. Cordero, T. Aydiello, R. A. Medina, K. Kisand, A. Puel, E. Jouanguy, L. Abel, A. Cobat, S. Trouillet-Assant, A. Garcia-Sastre, J. L. Casanova, Autoantibodies against type I IFNs in patients with critical influenza pneumonia. *J. Exp. Med.* **219**, e20220514 (2022).
 35. J. Manry, P. Bastard, A. Gervais, T. Le Voyer, J. Rosain, Q. Philippot, E. Michailidis, H. H. Hoffmann, S. Eto, M. Garcia-Prat, L. Bizien, A. Parra-Martinez, R. Yang, L. Haljasmagi, M. Migaud, K. Sarekannu, J. Maslovskaja, N. de Prost, Y. Tandjaoui-Lambiotte, C. E. Luyt, B. Amador-Borrero, A. Gaudet, J. Poissy, P. Morel, P. Richard, J. Cognasse, J. Troya, S. Trouillet-Assant, A. Belot, K. Saker, P. Garcon, J. G. Riviere, J. C. Lagier, S. Gentile, L. B. Rosen, E. Shaw, T. Morio, J. Tanaka, D. Dalmau, P. L. Tharoux, D. Sene, A. Stepanian, B. Megarbane, V. Triantafyllia, A. Fekkar, J. R. Heath, J. L. Franco, J. M. Anaya, J. Sole-Violan, L. Imberti, A. Biondi, P. Bonfanti, R. Castagnoli, O. M. Delmonte, Y. Zhang, A. L. Snow, S. M. Holland, C. M. Biggs, M. Moncada-Velez, A. A. Arias, L. Lorenzo, S. Boucherit, D. Anglicheau, A. M. Planas, F. Haerynck, S. Duvlis, T. Ozcelik, S. Keles, A. A. Bousfiha, J. El Bakkouri, C. Ramirez-Santana, S. Paul, Q. Pan-Hammarstrom, L. Hammarstrom, A. Dupont, A. Kurolap, C. N. Metz, A. Aiuti, G. Casari, V. Lampasona, F. Ciceri, L. A. Barreiros, E. Dominguez-Garrido, M. Vidigal, M. Zatz, D. van de Beek, S. Sahanic, I. Tancevski, Y. Stepanovskyy, O. Boyarchuk, Y. Nukui, M. Tsumura, L. Vidaur, S. G. Tangye, S. Burrel, D. Duffy, L. Quintana-Murci, A. Klocperk, N. Y. Kann, A. Shcherbina, Y. L. Lau, D. Leung, M. Coulangeat, J. Marlet, R. Koning, L. F. Reyes, A. Chauvineau-Grenier, F. Venet, G. Monneret, M. C. Nussenzweig, R. Arrestier, I. Boudhabhay, H. Baris-Feldman, D. Hagin, J. Wauters, I. Meyts, A. H. Dyer, S. P. Kennelly, N. M. Bourke, R. Halwani, F. S. Sharif-Askari, K. Dorgham, J. Sallette, S. M. Sedkaoui, S. AlKhater, R. Rigo-Bonnin, F. Morandeira, L. Roussel, D. C. Vinh, C. Erikstrup, A. Condino-Neto, C. Prando, A. Bondarenko, A. N. Spaan, L. Gardin, J. S. Lyonnet, K. Bilguvar, R. P. Lifton, S. Mane, HGID Lab, COVID Clinicians; COVID-STORM Clinicians, NIAID Immune Response to COVID Group, NH-COVAR Study Group, Danish CHGE; Danish Blood Donor Study, St. James's Hospital, SARS CoV2 Interest Group, French COVID Cohort Study Group, Imagine COVID-Group, Milieu Interieur Consortium, CoV-Contact Cohort, Amsterdam UMC Covid-19 Biobank Investigators, COVID Human Genetic Effort, CP-COVID-19 Group, CONSTANCES cohort, 3C-Dijon Study, Cerba Health-Care, Etablissement Français du Sang Study group, M. S. Anderson, B. Boisson, V. Beziat, S. Y. Zhang, E. Andreaskos, O. Hermine, A. Pujol, P. Peterson, T. H. Mogensen, L. Rowen, J. Mond, S. Debette, X. de Lamballerie, C. Burdet, L. Bouadma, M. Zins, P. Soler-Palacin, R. Colobran, G. Gorochov, X. Solanich, S. Susen, J. Martinez-Picado, D. Raoult, M. Vasse, P. K. Gregersen, L. Piemonti, C. Rodriguez-Gallego, L. D. Notarangelo, H. C. Su, K. Kisand, S. Okada, A. Puel, E. Jouanguy, C. M. Rice, P. Tiberghien, Q. Zhang, J. L. Casanova, L. Abel, A. Cobat, The risk of COVID-19 death is much greater and age dependent with type I IFN autoantibodies. *Proc. Natl. Acad. Sci. U.S.A.* **119**, e2200413119 (2022).
 36. F. Alotaibi, N. K. Alharbi, L. B. Rosen, A. Y. Asiri, H. H. Balkhy, M. Al Jeraisy, Y. Mandourah, S. AlJohani, S. Al Harbi, H. A. A. Jokhdar, A. M. Deeb, Z. A. Memish, J. Jose, S. Ghazal, S. Al Faraj, G. A. Al Mekhlafi, N. M. Sherbeen, F. E. Elzein, B. M. AlMutairi, A. Al-Dawood, M. L. Abdullah, T. Barhouni, M. W. Alenazi, A. Almasood, S. M. Holland, Y. M. Arabi, G., Type I interferon autoantibodies in hospitalized patients with Middle East respiratory syndrome and association with outcomes and treatment effect of interferon beta-1b in MIRACLE clinical trial. *Influenza Other Respir. Viruses* **17**, e13116 (2023).
 37. A. Gervais, F. Rovida, M. A. Avanzini, S. Croce, A. Marchal, S. C. Lin, A. Ferrari, C. W. Thorball, O. Constant, T. Le Voyer, Q. Philippot, J. Rosain, M. Angelini, M. Perez Lorenzo, L. Bizien, C. Achille, F. Trespidi, E. Burdino, I. Cassaniti, D. Lillier, C. Fornara, J. C. Sammartino, D. Cereda, C. Marrocu, A. Piralla, C. Valsecchi, S. Ricagno, P. Cogo, O. Neth, I. Marin-Cruz, M. Pacenti, A. Sinigaglia, M. Trevisan, A. Volpe, A. Marzollo, F. Conti, T. Lazzarotto, A. Pession, P. Viale, J. Fellay, S. Ghirardello, M. Aubart, V. Ghisetti, A. Aiuti, E. Jouanguy, P. Bastard, E. Percivalle, F. Baldanti, A. Puel, M. R. MacDonald, C. M. Rice, G. Rossini, K. O. Murray, Y. Simonin, A. Nagy, L. Barzon, L. Abel, M. S. Diamond, A. Cobat, S. Y. Zhang, J. L. Casanova, A. Borghesi, Autoantibodies neutralizing type I IFNs underlie West Nile virus encephalitis in ~40% of patients. *J. Exp. Med.* **220**, e20230661 (2023).
 38. P. Bastard, E. Michailidis, H. H. Hoffmann, M. Chbihi, T. Le Voyer, J. Rosain, Q. Philippot, Y. Seeleuthner, A. Gervais, M. Materna, P. M. N. de Oliveira, M. L. S. Maia, A. P. D. A. Bom, T. Azamor, D. A. da Conceicao, E. Goudouris, A. Homma, G. Slesak, J. Schafer, B. Pulendran, J. D. Miller, R. Huits, R. Yang, L. B. Rosen, L. Bizien, L. Lorenzo, M. Chrabieh, L. V. Erazo, F. Rozenberg, M. M. Jeljel, V. Beziat, S. M. Holland, A. Cobat, L. D. Notarangelo, H. C. Su, R. Ahmed, A. Puel, S. Y. Zhang, L. Abel, S. J. Seligman, Q. Zhang, M. R. MacDonald, E. Jouanguy, C. M. Rice, J. L. Casanova, Auto-antibodies to type I IFNs can underlie adverse reactions to yellow fever live attenuated vaccine. *J. Exp. Med.* **218**, (2021).
 39. E. S. Husebye, M. S. Anderson, O. Kampe, Autoimmune polyendocrine syndromes. *N. Engl. J. Med.* **378**, 1132–1141 (2018).
 40. T. Le Voyer, A. V. Parent, X. Liu, A. Cederholm, A. Gervais, J. Rosain, T. Nguyen, M. P. Lorenzo, E. Rackaityte, D. Rinchai, P. Zhang, L. Bizien, G. Hancioglu, P. Ghillani-Dalbin, J. L. Charuel, Q. Philippot, M. S. Gueye, M. L. M. Renkilaraj, M. Ogishi, C. Soude, M. Migaud, F. Rozenberg, M. Momenilandi, Q. Riller, L. Imberti, O. M. Delmonte, G. Muller, B. Keller, J. Orrego, W. A. F. Gallego, T. Rubin, M. Emiroglu, N. Parvaneh, D. Eriksson, M. Aranda-Guillen, D. I. Berrios, L. Vong, C. H. Katelaris, P. Mustillo, J. Raedler, J. Bohlen, J. B. Celik, C. Astudillo, S. Winter, NF- κ B Consortium, COVID Human Genetic Effort, C. McLean, A. Guffroy, J. L. Derisi, D. Yu, C. Miller, Y. Feng, A. Guichard, V. Beziat, J. Bustamante, Q. Pan-Hammarstrom, Y. Zhang, L. B. Rosen, S. M. Holland, M. Bosticardo, H. Kenney, R. Castagnoli, C. A. Slade, K. Bostuz, N. Mahlaoui, S. Latour, R. S. Abraham, V. Lougaris, F. Hauck, A. Sediva, F. Atschekzei, G. Sogkas, M. C. Poli, M. A. Slatter, B. Palterer, M. D. Keller, A. Pinzon-Charry, A. Sullivan, L. Droney, D. Suan, M. Wong, A. Kane, H. Hu, C. Ma, H. Grombrikova, P. Ciznar, I. Dalal, N. Aladjidi, M. Hie, E. Lazaro, J. Franco, S. Keles, M. Malphettes, M. Pasquet, M. E. Maccari, A. Meinhardt, A. Ikinciogullari, M. Shahrooi, F. Celme, P. Frosk, C. C. Goodnow, P. E. Gray, A. Belot, H. S. Kuehn, S. D. Rosenzweig, M. Miyara, F. Licciardi, A. Servettaz, V. Barlogis, G. Le Guenno, V. M. Herrmann, T. Kuipers, G. Ducoux, F. Sarrot-Reynaud, C. Schuetz, C. Cunningham-Rundles, F. Rieux-Laucat, S. G. Tangye, C. Sobacchi, R. Doffinger, K. Warnatz, B. Grimbacher, C. Fieschi, L. Berteloot, V. L. Bryant, S. T. Assant, H. Su, B. Neven, L. Abel, Q. Zhang, B. Boisson, A. Cobat, E. Jouanguy, O. Kampe, P. Bastard, C. M. Roifman, N. Landegren, L. D. Notarangelo, M. S. Anderson, J. L. Casanova, A. Puel, Autoantibodies against type I IFNs in humans with alternative NF- κ B pathway deficiency. *Nature* **623**, 803–813 (2023).
 41. T. M. Campbell, Z. Liu, Q. Zhang, M. Moncada-Velez, L. E. Covill, P. Zhang, I. Alavi Darazam, P. Bastard, L. Bizien, G. Bucciol, S. Lind Enoksson, E. Jouanguy, S. N. Karabela, T. Khan, Y. Kendir-Demirkol, A. A. Arias, D. Mansouri, P. Marits, M. Marr, I. Migeotte, L. Moens, Z. Ozcelik, I. Pellier, A. Sendel, S. Senoglu, M. Shahrooi, C. I. E. Smith, I. Vandernoot, K. Willekens, K. Kart Yasar, C. H. G. Effort, P. Bergman, L. Abel, A. Cobat, J. L. Casanova, I. Meyts, Y. T. Bryceson, Correction: Respiratory viral infections in otherwise healthy humans with inherited IRF7 deficiency. *J. Exp. Med.* **219**, e20220202 (2022).
 42. M. J. Ciancanelli, S. X. Huang, P. Luthra, H. Garner, Y. Itan, S. Volpi, F. G. Lafaille, C. Trouillet, M. Schmolke, R. A. Albrecht, E. Israelsson, H. K. Lim, M. Casadio, T. Hermesh, L. Lorenzo, L. W. Leung, V. Pedergnana, B. Boisson, S. Okada, C. Picard, B. Ringier, F. Troussier, D. Chaussabel, L. Abel, I. Pellier, L. D. Notarangelo, A. Garcia-Sastre, C. F. Basler, F. Geissmann, S. Y. Zhang, H. W. Snoeck, J. L. Casanova, Infectious disease. Life-threatening influenza and impaired interferon amplification in human IRF7 deficiency. *Science* **348**, 448–453 (2015).
 43. X. Zhang, D. Bogunovic, B. Payelle-Brogard, V. Francois-Newton, S. D. Speer, C. Yuan, S. Volpi, Z. Li, O. Sanal, D. Mansouri, I. Tezcan, G. I. Rice, C. Chen, N. Mansouri, S. A. Mahdavian, Y. Itan, B. Boisson, S. Okada, L. Zeng, X. Wang, H. Jiang, W. Liu, T. Han, D. Liu, T. Ma, B. Wang, M. Liu, J. Y. Liu, Q. K. Wang, D. Yalmozoglu, L. Radoshevich, G. Uze, P. Gros, F. Rozenberg, S. Y. Zhang, E. Jouanguy, J. Bustamante, A. Garcia-Sastre, L. Abel, P. Lebon, L. D. Notarangelo, Y. J. Crow, S. Boisson-Dupuis, J. L. Casanova, S. Pellegrini, Human intracellular ISG15 prevents interferon- α/β over-amplification and auto-inflammation. *Nature* **517**, 89–93 (2015).
 44. D. Lee, J. Le Pen, A. Yatim, B. Dong, Y. Aquino, M. Ogishi, R. Pescarmona, E. Talouarn, D. Rinchai, P. Zhang, M. Perret, Z. Liu, I. Jordan, S. E. Bozdemir, G. I. Bayhan, C. Beaufils, L. Bizien, A. Bisiaux, W. Lei, M. Hasan, J. Chen, C. Gaughan, A. Asthana, V. Libri, J. M. Luna, F. Jaffre, H. H. Hoffmann, E. Michailidis, M. Morews, Y. Seeleuthner, K. Bilguvar, S. Mane, C. Flores, Y. Zhang, A. A. Arias, R. Bailey, A. Schluter, B. Milisavljevic, B. Bigio, T. Le Voyer, M. Materna, A. Gervais, M. Moncada-Velez, F. Pala, T. Lazarov, R. Levy, A. L. Neehus, J. Rosain, J. Peel, Y. H. Chan, M. P. Morin, R. M. Pino-Ramirez, S. Belkaya, L. Lorenzo, J. Anton, S. Delafontaine, J. Toubiana, F. Bajolle, V. Fumado, M. L. De Diego, N. Fidouch, F. Rozenberg, J. Perez-Tur, S. Chen, T. Evans, F. Geissmann, P. Lebon, S. R. Weiss, D. Bonnet, X. Duval, CoV-Contact Cohort, COVID Human Genetic Effort, Q. Pan-Hammarstrom, A. M. Planas, I. Meyts, F. Haerynck, A. Pujol, V. Sancho-Shimizu, C. L. Dalgard, J. Bustamante, A. Puel, S. Boisson-Dupuis, B. Boisson, T. Maniatis, Q. Zhang, P. Bastard, L. Notarangelo, V. Beziat, R. P. de Diego, C. Rodriguez-Gallego, H. C. Su, R. P. Lifton, E. Jouanguy, A. Cobat, L. Alsina, S. Keles, E. Haddad, L. Abel, A. Belot, L. Quintana-Murci, C. M. Rice, R. H. Silverman, S. Y. Zhang, J. L. Casanova, Inborn errors of OAS-RNase L in

- SARS-CoV-2-related multisystem inflammatory syndrome in children. *Science* **379**, eabo3627 (2023).
45. M. E. Meuwissen, R. Schot, S. Buta, G. Oudesluijs, S. Tinschert, S. D. Speer, Z. Li, L. van Unen, D. Heijnsman, T. Goldmann, M. H. Lequin, J. M. Kros, W. Stam, M. Hermann, R. Willemsen, R. W. Brouwer, I. W. F. Van, M. Martin-Fernandez, I. de Coo, J. Dudink, F. A. de Vries, A. Bertoli Avella, M. Prinz, Y. J. Crow, F. W. Verheijen, S. Pellegrini, D. Bogunovic, G. M. Mancini, Human USP18 deficiency underlies type 1 interferonopathy leading to severe pseudo-TORCH syndrome. *J. Exp. Med.* **213**, 1163–1174 (2016).
 46. C. J. A. Duncan, B. J. Thompson, R. Chen, G. I. Rice, F. Gothe, D. F. Young, S. C. Lovell, V. G. Shuttleworth, V. Brocklebank, B. Corner, A. J. Skelton, V. Bondet, J. Coxhead, D. Duffy, C. Fourrage, J. H. Livingston, J. Pavaine, E. Cheesman, S. Bitetti, A. Grainger, M. Acres, B. A. Innes, A. Mikulasova, R. Sun, R. Hussain, R. Wright, R. Wynn, M. Zarhrate, L. A. H. Zeef, K. Wood, S. M. Hughes, C. L. Harris, K. R. Engelhardt, Y. J. Crow, R. E. Randall, D. Kavanagh, S. Hambleton, T. A. Briggs, Severe type I interferonopathy and unrestrained interferon signalling due to a homozygous germline mutation in STAT2. *Sci. Immunol.* **4**, eaav7501 (2019).
 47. C. Gruber, M. Martin-Fernandez, F. Ailal, X. Qiu, J. Taft, J. Altman, J. Rosain, S. Buta, A. Bousfiha, J. L. Casanova, J. Bustamante, D. Bogunovic, Homozygous STAT2 gain-of-function mutation by loss of USP18 deficiency in a patient with type I interferonopathy. *J. Exp. Med.* **217**, e20192319 (2020).
 48. Y. Chen, L. Graf, T. Chen, Q. Liao, T. Bai, P. P. Petric, W. Zhu, L. Yang, J. Dong, J. Lu, Y. Chen, J. Shen, O. Haller, P. Staeheli, G. Kochs, D. Wang, M. Schwemmle, Y. Shu, Rare variant MX1 alleles increase human susceptibility to zoonotic H7N9 influenza virus. *Science* **373**, 918–922 (2021).
 49. Y. Crow, Categorisation of monogenic disorders: Underlying a deficit of type I IFN activity / observed type I IFN signalling upregulation. Dryad (2024); <https://doi.org/10.5061/dryad.1zcrjdg1c>.
 50. S. Y. Zhang, E. Jouanguy, S. Ugolini, A. Smahi, G. Elain, P. Romero, D. Segal, V. Sancho-Shimizu, L. Lorenzo, A. Puel, C. Picard, A. Chappier, S. Planoulaine, M. Titeux, C. Cognet, H. von Bernuth, C. L. Ku, A. Casrouge, X. X. Zhang, L. Barreiro, J. Leonard, C. Hamilton, P. Lebon, B. Heron, L. Vallee, L. Quintana-Murci, A. Hovnanian, F. Rozenberg, E. Vivier, F. Geissmann, M. Tardieu, L. Abel, J. L. Casanova, TLR3 deficiency in patients with herpes simplex encephalitis. *Science* **317**, 1522–1527 (2007).
 51. H. K. Lim, S. X. L. Huang, J. Chen, G. Kerner, O. Gilliaux, P. Bastard, K. Dobbs, N. Hernandez, N. Goudin, M. L. Hasek, E. J. Garcia Reino, F. G. Lafaille, L. Lorenzo, P. Luthra, T. Kochetkov, B. Bigio, S. Boucherit, F. Rozenberg, C. Vedrinne, M. D. Keller, Y. Itan, A. Garcia-Sastre, M. Celard, J. S. Orange, M. J. Ciancanelli, I. Meyts, Q. Zhang, L. Abel, L. D. Notarangelo, H. W. Snoeck, J. L. Casanova, S. Y. Zhang, Severe influenza pneumonitis in children with inherited TLR3 deficiency. *J. Exp. Med.* **216**, 2038–2056 (2019).
 52. D. Gao, M. J. Ciancanelli, P. Zhang, O. Harschnitz, V. Bondet, M. Hasek, J. Chen, X. Mu, Y. Itan, A. Cobat, V. Sancho-Shimizu, B. Bigio, L. Lorenzo, G. Ciceri, J. McAlpine, E. Anguiano, E. Jouanguy, D. Chaussabel, I. Meyts, M. S. Diamond, L. Abel, S. Hur, G. A. Smith, L. Notarangelo, D. Duffy, L. Studer, J. L. Casanova, S. Y. Zhang, TLR3 controls constitutive IFN- β antiviral immunity in human fibroblasts and cortical neurons. *J. Clin. Invest.* **131**, e134529 (2021).
 53. T. Asano, B. Boisson, F. Onodi, D. Matuozzo, M. Moncada-Velez, M. R. L. Maglorius Renkilaraj, P. Zhang, L. Meertens, A. Bolze, M. Materna, S. Korniotis, A. Gervais, E. Talouarn, B. Bigio, Y. Seeleuthner, K. Bilguvar, Y. Zhang, A. L. Neehus, M. Ogishi, S. J. Pelham, T. Le Voyer, J. Rosain, Q. Philippot, P. Soler-Palacin, R. Colobran, A. Martin-Nalda, J. G. Riviere, Y. Tandjaoui-Lambiotte, K. Chaibi, M. Shahrooei, I. A. Darazam, N. A. Olyaei, D. Mansouri, N. Hatipoglu, F. Palabiyik, T. Ozcelik, G. Novelli, A. Novelli, G. Casari, A. Aiuti, P. Carrera, S. Bondesan, F. Barzaghi, P. Rovere-Querini, C. Tresoldi, J. L. Franco, J. Rojas, L. F. Reyes, I. G. Bustos, A. A. Arias, G. Morelle, K. Christele, J. Troya, L. Planas-Serra, A. Schluter, M. Gut, A. Pujol, L. M. Allende, C. Rodriguez-Gallego, C. Flores, O. Cabrera-Marante, D. Duffy, L. E. Pleguezuelo, R. P. de Diego, S. Keles, G. Aytekin, O. M. Akcan, Y. T. Bryceson, P. Bergman, P. Brodin, D. Smole, C. I. E. Smith, A. C. Norlin, T. M. Campbell, L. E. Covill, L. Hammarstrom, Q. Pan-Hammarstrom, H. Abolhassani, S. Mane, N. Marr, M. Ata, F. Al Ali, T. Khan, A. N. Spaan, C. L. Dalgard, P. Bonfanti, A. Biondi, S. Tubiana, C. Burdet, R. Nussbaum, A. Kahn-Kirby, A. L. Snow, COVID Human Genetic Effort, COVID-STORM Clinicians, COVID Clinicians; Imagine COVID Group, French COVID Cohort Study Group, CoV-Contact Cohort; Amsterdam UMC Covid, Biobank, NIAID-USUHS COVID Study Group, J. Bustamante, A. Puel, S. Boisson-Dupuis, S. Y. Zhang, V. Beziat, R. P. Lifton, L. D. Notarangelo, L. Abel, H. C. Su, E. Jouanguy, A. Amara, V. Soumelis, A. Cobat, Q. Zhang, J. L. Casanova, X-linked recessive TLR7 deficiency in ~1% of men under 60 years old with life-threatening COVID-19. *Sci. Immunol.* **6**, eabl4348 (2021).
 54. F. Onodi, L. Bonnet-Madin, L. Meertens, L. Karpf, J. Poirot, S. Y. Zhang, C. Picard, A. Puel, E. Jouanguy, Q. Zhang, J. Le Goff, J. M. Molina, C. Delaugerre, J. L. Casanova, A. Amara, V. Soumelis, SARS-CoV-2 induces human plasmacytoid predendritic cell diversification via UNC93B and IRAK4. *J. Exp. Med.* **218**, e20201387 (2021).
 55. A. Garcia-Garcia, R. Perez de Diego, C. Flores, D. Rinchai, J. Sole-Violan, A. Deya-Martinez, B. Garcia-Solis, J. M. Lorenzo-Salazar, E. Hernandez-Brito, A. L. Lanz, L. Moens, G. Bucciol, M. Almuqamam, J. B. Domachowski, E. Colino, J. L. Santos-Perez, F. M. Marco, C. Pignata, A. Bousfiha, S. E. Turvey, S. Bauer, F. Haerynck, J. G. Ocejo-Vinyals, F. Lendinez, S. Prader, N. Naumann-Bartsch, J. Pachlopnik Schmid, C. M. Biggs, K. Hildebrand, A. Dreesman, M. A. Cardenes, F. Ailal, I. Benhsaien, G. Giardino, A. Molina-Fuentes, C. Fortuny, S. Madhavarapu, D. H. Conway, C. Prando, L. Schildowski, M. T. M. de Saavedra Alvarez, R. Alfaro, F. R. de Castro, ESID Registry Working Party, COVID Human Genetic Effort, I. Meyts, F. Hauck, A. Puel, P. Bastard, B. Boisson, E. Jouanguy, L. Abel, A. Cobat, Q. Zhang, J. L. Casanova, L. Alsina, C. Rodriguez-Gallego, Humans with inherited MyD88 and IRAK-4 deficiencies are predisposed to hypoxemic COVID-19 pneumonia. *J. Exp. Med.* **220**, e20220170 (2023).
 56. A. Casrouge, S. Y. Zhang, C. Eidschinken, E. Jouanguy, A. Puel, K. Yang, A. Alcais, C. Picard, N. Mahfoufi, N. Nicolas, L. Lorenzo, S. Planoulaine, B. Senechal, F. Geissmann, K. Tabeta, K. Hoebe, X. Du, R. L. Miller, B. Heron, C. Mignot, T. B. de Villemeur, P. Lebon, O. Dulac, F. Rozenberg, B. Beutler, M. Tardieu, L. Abel, J. L. Casanova, Herpes simplex virus encephalitis in human UNC-93B deficiency. *Science* **314**, 308–312 (2006).
 57. I. T. Lamborn, H. Jing, Y. Zhang, S. B. Drutman, J. K. Abbott, S. Munir, S. Bade, H. M. Murdock, C. P. Santos, L. G. Brock, E. Masutani, E. Y. Fordjour, J. J. McElwee, J. D. Hughes, D. P. Nichols, A. Belkadi, A. J. Oler, C. S. H. Appel, H. F. Matthews, L. Abel, P. L. Collins, K. Subbarao, E. W. Gelfand, M. J. Ciancanelli, J. L. Casanova, H. C. Su, Recurrent rhinovirus infections in a child with inherited MDA5 deficiency. *J. Exp. Med.* **214**, 1949–1972 (2017).
 58. S. Asgari, L. J. Schlappbach, S. Anchisi, C. Hammer, I. Bartha, T. Junier, G. Mottet-Osman, K. M. Posfay-Barbe, D. Longchamp, M. Stocker, S. Cordey, L. Kaiser, T. Riedel, T. Kenna, D. Long, A. Schibler, A. Telenti, C. Tapparel, P. J. McLaren, D. Garcin, J. Fellay, Severe viral respiratory infections in children with IFIH1 loss-of-function mutations. *Proc. Natl. Acad. Sci. U.S.A.* **114**, 8342–8347 (2017).
 59. J. Chen, H. Jing, A. Martin-Nalda, P. Bastard, J. G. Riviere, Z. Liu, R. Colobran, D. Lee, W. Tung, J. Manry, M. Hasek, S. Boucherit, L. Lorenzo, F. Rozenberg, M. Aubart, L. Abel, H. C. Su, P. Soler Palacin, J. L. Casanova, S. Y. Zhang, Inborn errors of TLR3- or MDA5-dependent type I IFN immunity in children with enterovirus rhombencephalitis. *J. Exp. Med.* **218**, e20211349 (2021).
 60. B. Ogunjimi, S. Y. Zhang, K. B. Sorensen, K. A. Skipper, M. Carter-Timofte, G. Kerner, S. Luecke, T. Prabakaran, Y. Cai, J. Meester, E. Bartholomeus, N. A. Bolar, G. Vandeweyer, C. Claes, Y. Sillis, L. Lorenzo, R. A. Fiorenza, S. Boucherit, C. Dielman, S. Heynderickx, G. Elias, A. Kurotova, A. V. Auwera, L. Verstraete, L. Lagae, H. Verhelst, A. Jansen, J. Ramet, A. Suls, E. Smits, B. Ceulemans, L. Van Laer, G. Plat Wilson, J. Kreth, C. Picard, H. Von Bernuth, J. Fluss, S. Chabrier, L. Abel, G. Mortier, S. Fribourg, J. G. Mikkelsen, J. L. Casanova, S. R. Paludan, T. H. Mogensen, Inborn errors in RNA polymerase III underlie severe varicella zoster virus infections. *J. Clin. Invest.* **127**, 3543–3556 (2017).
 61. M. Herman, M. Ciancanelli, Y. H. Ou, L. Lorenzo, M. Klauedel-Dreszler, E. Pauwels, V. Sancho-Shimizu, R. Perez de Diego, A. Abhyankar, E. Israelsson, Y. Guo, A. Cardon, F. Rozenberg, P. Lebon, M. Tardieu, E. Heropolitanska-Pliszka, D. Chaussabel, M. A. White, L. Abel, S. Y. Zhang, J. L. Casanova, Heterozygous TBK1 mutations impair TLR3 immunity and underlie herpes simplex encephalitis of childhood. *J. Exp. Med.* **209**, 1567–1582 (2012).
 62. L. L. Andersen, N. Mork, L. S. Reinert, E. Kofod-Olsen, R. Narita, S. E. Jorgensen, K. A. Skipper, K. Honing, H. H. Gad, L. Ostergaard, T. F. Orntoft, V. Hornung, S. R. Paludan, J. G. Mikkelsen, T. Fujita, M. Christiansen, R. Hartmann, T. H. Mogensen, Functional IRF3 deficiency in a patient with herpes simplex encephalitis. *J. Exp. Med.* **212**, 1371–1379 (2015).
 63. I. Gresser, L. Morel-Maroger, Y. Riviere, J. C. Guillon, M. G. Tovey, D. Woodrow, J. C. Sloper, J. Moss, Interferon-induced disease in mice and rats. *Ann. N. Y. Acad. Sci.* **350**, 12–20 (1980).
 64. P. Lebon, J. Badoual, G. Ponsot, F. Goutieres, F. Hemeury-Cukier, J. Aicardi, Intrathecal synthesis of interferon-alpha in infants with progressive familial encephalopathy. *J. Neurol. Sci.* **84**, 201–208 (1988).
 65. J. Aicardi, F. Goutieres, A progressive familial encephalopathy in infancy with calcifications of the basal ganglia and chronic cerebrospinal fluid lymphocytosis. *Ann. Neurol.* **15**, 49–54 (1984).
 66. T. Vesikari, A. Nuutila, K. Cantell, Neurologic sequelae following interferon therapy of juvenile laryngeal papilloma. *Acta Paediatr. Scand.* **77**, 619–622 (1988).
 67. L. E. Ronnblom, G. V. Alm, K. E. Oberg, Possible induction of systemic lupus erythematosus by interferon-alpha treatment in a patient with a malignant carcinoid tumour. *J. Intern. Med.* **227**, 207–210 (1990).
 68. A. Tolaymat, B. Leventhal, A. Sakarcin, H. Kashima, C. Monteiro, Systemic lupus erythematosus in a child receiving long-term interferon therapy. *J. Pediatr.* **120**, 429–432 (1992).
 69. H. Smedley, M. Katrak, K. Sikora, T. Wheeler, Neurological effects of recombinant human interferon. *Br. Med. J. (Clin. Res. Ed)* **286**, 262–264 (1983).
 70. Y. Akwa, D. E. Hassett, M. L. Eloranta, K. Sandberg, E. Masliah, H. Powell, J. L. Whitton, F. E. Bloom, I. L. Campbell, Transgenic expression of IFN-alpha in the central nervous system of mice protects against lethal neurotropic viral infection but induces inflammation and neurodegeneration. *J. Immunol.* **161**, 5016–5026 (1998).

71. Y. J. Crow, B. E. Hayward, R. Parmar, P. Robins, A. Leitch, M. Ali, D. N. Black, H. van Bokhoven, H. G. Brunner, B. C. Hamel, P. C. Corry, F. M. Cowan, S. G. Frints, J. Klepper, J. H. Livingston, S. A. Lynch, J. F. Massey, J. F. Meritet, J. L. Michaud, G. Ponsot, T. Voit, P. Lebon, D. T. Bonthron, A. P. Jackson, D. E. Barnes, T. Lindahl, Mutations in the gene encoding the 3'-5' DNA exonuclease TREX1 cause Aicardi-Goutières syndrome at the AGS1 locus. *Nat. Genet.* **38**, 917–920 (2006).
72. Y. J. Crow, A. Leitch, B. E. Hayward, A. Garner, R. Parmar, E. Griffith, M. Ali, C. Sempile, J. Aicardi, R. Babul-Hirji, C. Baumann, P. Baxter, E. Bertini, K. E. Chandler, D. Chitayat, D. Cau, C. Dery, E. Fazzi, C. Goizet, M. D. King, J. Klepper, D. Lacombe, G. Lanzi, H. Lyall, M. L. Martinez-Frias, M. Mathieu, C. McKeown, A. Monier, Y. Oade, O. W. Quarrell, C. D. Ritter, R. C. Rogers, A. Sanchis, J. B. Stephenson, U. Tacke, M. Till, J. L. Tolmie, P. Tomlin, T. Voit, B. Weschke, C. G. Woods, P. Lebon, D. T. Bonthron, C. P. Ponting, A. P. Jackson, Mutations in genes encoding ribonuclease H2 subunits cause Aicardi-Goutières syndrome and mimic congenital viral brain infection. *Nat. Genet.* **38**, 910–916 (2006).
73. G. I. Rice, J. Bond, A. Asipu, R. L. Brunette, I. W. Manfield, I. M. Carr, J. C. Fuller, R. M. Jackson, T. Lamb, T. A. Briggs, M. Ali, H. Gornall, L. R. Couthard, A. Aeby, S. P. Attard-Montalto, E. Bertini, C. Bodemer, K. Brockmann, L. A. Brueton, P. C. Corry, I. Desguerre, E. Fazzi, A. G. Cazorla, B. Gener, B. C. Hamel, A. Heiberg, M. Hunter, M. S. van der Knaap, R. Kumar, L. Lagae, P. G. Landrieu, C. M. Lourenco, D. Marom, M. F. McDermott, W. van der Merwe, S. Orcesi, J. S. Prendiville, M. Rasmussen, S. A. Shalev, D. M. Soler, M. Shinawi, R. Spiegel, T. Y. Tan, A. Vanderver, E. L. Wakeling, E. Wassmer, E. Whittaker, P. Lebon, D. B. Stetson, D. T. Bonthron, Y. J. Crow, Mutations involved in Aicardi-Goutières syndrome implicate SAMHD1 as regulator of the innate immune response. *Nat. Genet.* **41**, 829–832 (2009).
74. G. I. Rice, P. R. Kasher, G. M. Forte, N. M. Mannion, S. M. Greenwood, M. Szykiewicz, J. E. Dickerson, S. S. Bhaskar, M. Zampini, T. A. Briggs, E. M. Jenkinson, C. A. Bacino, R. Battini, E. Bertini, P. A. Brogan, L. A. Brueton, M. Carpanelli, C. De Laet, P. de Lonlay, M. del Toro, I. Desguerre, E. Fazzi, A. Garcia-Cazorla, A. Heiberg, M. Kawaguchi, R. Kumar, J. P. Lin, C. M. Lourenco, A. M. Male, W. Marques Jr., C. Mignot, I. Olivieri, S. Orcesi, P. Prabhakar, M. Rasmussen, R. A. Robinson, F. Rozenberg, J. L. Schmidt, K. Steindl, T. Y. Tan, W. G. van der Merwe, A. Vanderver, G. Vassallo, E. L. Wakeling, E. Wassmer, E. Whittaker, J. H. Livingston, P. Lebon, T. Suzuki, P. J. McLaughlin, L. P. Keegan, M. A. O'Connell, S. C. Lovell, Y. J. Crow, Mutations in ADAR1 cause Aicardi-Goutières syndrome associated with a type I interferon signature. *Nat. Genet.* **44**, 1243–1248 (2012).
75. C. Ugenti, A. Lepellet, M. Depp, A. P. Badrock, M. P. Rodero, M. T. El-Daher, G. I. Rice, S. Dhir, A. P. Wheeler, A. Dhir, W. Albawardi, M. L. Fremond, L. Seabra, J. Doig, N. Blair, M. J. Martin-Niclos, E. D. Mina, A. Rubio-Roldan, J. L. Garcia-Perez, D. Sproul, J. Rehwinkel, J. Hertzog, A. Boland-Auge, R. Olaso, J. F. Deleuze, J. Baruteau, K. Brochard, J. Buckley, V. Cavallera, C. Cereda, L. M. H. De Waele, A. Dobbie, D. Doummar, F. Elmslie, M. Koch-Hogrebe, R. Kumar, K. Lamb, J. H. Livingston, A. Majumdar, C. M. Lorenzo, S. Orcesi, S. Peudenier, K. Rostasy, C. A. Salmon, C. Scott, D. Tonduti, G. Touati, M. Valente, H. van der Linden Jr., H. Van Esch, M. Vermelle, K. Webb, A. P. Jackson, M. A. M. Reijns, N. Gilbert, Y. J. Crow, cGAS-mediated induction of type I interferon due to inborn errors of histone pre-mRNA processing. *Nat. Genet.* **52**, 1364–1372 (2020).
76. G. I. Rice, Y. Del Toro Duany, E. M. Jenkinson, G. M. Forte, B. H. Anderson, G. Ariaudo, B. Bader-Meunier, E. M. Baildam, R. Battini, M. W. Beresford, M. Casarano, M. Chouchane, R. Cimaz, A. E. Collins, N. J. Cordeiro, R. C. Dale, J. E. Davidson, L. De Waele, I. Desguerre, L. Faivre, E. Fazzi, B. Isidor, L. Lagae, A. R. Latchman, P. Lebon, C. Li, J. H. Livingston, C. M. Lourenco, M. M. Mancardi, A. Masurel-Paulet, I. B. McInnes, M. P. Menezes, C. Mignot, J. O'Sullivan, S. Orcesi, P. P. Picco, E. Riva, R. A. Robinson, D. Rodriguez, E. Salvatici, C. Scott, M. Szybowska, J. L. Tolmie, A. Vanderver, C. Vanhulle, J. P. Vieira, K. Webb, R. N. Whitney, S. G. Williams, L. A. Wolfe, S. M. Zuberi, S. Hur, Y. J. Crow, Gain-of-function mutations in IFIH1 cause a spectrum of human disease phenotypes associated with upregulated type I interferon signaling. *Nat. Genet.* **46**, 503–509 (2014).
77. Y. J. Crow, Type I interferonopathies: A novel set of inborn errors of immunity. *Ann. N. Y. Acad. Sci.* **1238**, 91–98 (2011).
78. K. D. Sullivan, H. C. Lewis, A. A. Hill, A. Pandey, L. P. Jackson, J. M. Cabral, K. P. Smith, L. A. Liggett, E. B. Gomez, M. D. Galbraith, J. DeGregori, J. M. Espinosa, Trisomy 21 consistently activates the interferon response. *eLife* **5**, e16220 (2016).
79. X. F. Kong, L. Worley, D. Rinchai, P. V. Jithesh, M. Goulet, E. Nonnotte, A. S. Rebillet, M. Conte, C. Mircher, N. Gurtler, L. Liu, M. Migaud, M. Elanbari, T. Habib, C. S. Ma, J. Bustamante, L. Abel, A. Ravel, S. Lyonnet, A. Munnich, D. Duffy, D. Chaussabel, J. L. Casanova, S. G. Tangye, S. Boisson-Dupuis, A. Puel, Three copies of four interferon receptor genes underlie a mild type I interferonopathy in down syndrome. *J. Clin. Immunol.* **40**, 807–819 (2020).
80. L. Malle, M. Martin-Fernandez, S. Buta, A. Richardson, D. Bush, D. Bogunovic, Excessive negative regulation of type I interferon disrupts viral control in individuals with Down syndrome. *Immunity* **55**, 2074–2084.e5 (2022).
81. K. A. Waugh, R. Minter, J. Baxter, C. Chi, M. D. Galbraith, K. D. Tuttle, N. P. Eduthan, K. T. Kinning, Z. Andrysik, P. Araya, H. Dougherty, L. N. Dunn, M. Ludwig, K. A. Schade, D. Tracy, K. P. Smith, R. E. Granrath, N. Busquet, S. Khanal, R. D. Anderson, L. L. Cox, B. E. Estrada, A. L. Rachubinski, H. R. Lyford, E. C. Britton, K. A. Fantauzzo, D. J. Orlicky, J. L. Matsuda, K. Song, T. C. Cox, K. D. Sullivan, J. M. Espinosa, Triplication of the interferon receptor locus contributes to hallmarks of Down syndrome in a mouse model. *Nat. Genet.* **55**, 1034–1047 (2023).
82. M. D. Galbraith, A. L. Rachubinski, K. P. Smith, P. Araya, K. A. Waugh, B. Enriquez-Estrada, K. Worek, R. E. Granrath, K. T. Kinning, N. Paul Eduthan, M. P. Ludwig, E. W. Y. Hsieh, K. D. Sullivan, J. M. Espinosa, Multidimensional definition of the interferonopathy of Down syndrome and its response to JAK inhibition. *Sci. Adv.* **9**, eadg6218 (2023).
83. B. Lin, R. Goldbach-Mansky, Pathogenic insights from genetic causes of autoinflammatory inflammasomopathies and interferonopathies. *J. Allergy Clin. Immunol.* **149**, 819–832 (2022).
84. E. Bartok, G. Hartmann, Immune sensing mechanisms that discriminate self from altered self and foreign nucleic acids. *Immunity* **53**, 54–77 (2020).
85. Y. J. Crow, D. B. Stetson, The type I interferonopathies: 10 years on. *Nat. Rev. Immunol.* **22**, 471–483 (2022).
86. A. Roers, B. Hiller, V. Hornung, Recognition of endogenous nucleic acids by the innate immune system. *Immunity* **44**, 739–754 (2016).
87. M. A. Lee-Kirsch, The type I interferonopathies. *Annu. Rev. Med.* **68**, 297–315 (2017).
88. K. Manthiram, Q. Zhou, I. Aksentijevich, D. L. Kastner, The monogenic autoinflammatory diseases define new pathways in human innate immunity and inflammation. *Nat. Immunol.* **18**, 832–842 (2017).
89. L. D. Notarangelo, R. Bacchetta, J. L. Casanova, H. C. Su, Human inborn errors of immunity: An expanding universe. *Sci. Immunol.* **5**, eabb1662 (2020).
90. D. B. Beck, A. Werner, D. L. Kastner, I. Aksentijevich, Disorders of ubiquitylation: Uncharted inflammation. *Nat. Rev. Rheumatol.* **18**, 435–447 (2022).
91. T. Saitoh, M. Yamamoto, M. Miyagishi, K. Taira, M. Nakanishi, T. Fujita, S. Akira, N. Yamamoto, S. Yamaoka, A20 is a negative regulator of IFN regulatory factor 3 signaling. *J. Immunol.* **174**, 1507–1512 (2005).
92. D. M. Schwartz, S. A. Blackstone, N. Sampaio-Moura, S. Rosenzweig, A. M. Burma, D. Stone, P. Hoffmann, A. Jones, T. Romeo, K. S. Barron, M. A. Waldman, I. Aksentijevich, D. L. Kastner, J. D. Milner, A. K. Ombrello, Type I interferon signature predicts response to JAK inhibition in haploinsufficiency of A20. *Ann. Rheum. Dis.* **79**, 429–431 (2020).
93. T. Miyamoto, Y. Honda, K. Izawa, N. Kanazawa, S. Kadowaki, H. Ohnishi, M. Fujimoto, N. Kambe, N. Kase, T. Shiba, Y. Nakagishi, S. Akizuki, K. Murakami, M. Bamba, Y. Nishida, A. Inui, T. Fujisawa, D. Nishida, N. Iwata, Y. Otsubo, S. Ishimori, M. Nishikori, K. Tanizawa, T. Nakamura, T. Ueda, Y. Ohwada, Y. Tsuyusaki, M. Shimizu, T. Ebato, K. Iwao, A. Kubo, T. Kawai, T. Matsubayashi, T. Miyazaki, T. Kanayama, M. Nishitani-Isa, H. Nihira, J. Abe, T. Tanaka, E. Hiejima, S. Okada, O. Ohara, M. K. Saito, J. Takita, R. Nishikomori, T. Yasumi, Assessment of type I interferon signatures in undifferentiated inflammatory diseases: A Japanese multicenter experience. *Front. Immunol.* **13**, 905960 (2022).
94. C. Zhang, X. Han, L. Sun, S. Yang, J. Peng, Y. Chen, Y. Jin, F. Xu, Z. Liu, Q. Zhou, Novel loss-of-function mutations in TNFAIP3 gene in patients with lupus nephritis. *Clin. Kidney J.* **15**, 2027–2038 (2022).
95. H. Oda, D. B. Beck, H. S. Kuehn, N. Sampaio Moura, P. Hoffmann, M. Ibarra, J. Stoddard, W. L. Tsai, G. Gutierrez-Cruz, M. Gadina, S. D. Rosenzweig, D. L. Kastner, L. D. Notarangelo, I. Aksentijevich, Second case of HOIP deficiency expands clinical features and defines inflammatory transcriptome regulated by LUBAC. *Front. Immunol.* **10**, 479 (2019).
96. P. Tao, S. Wang, S. Ozen, P. Y. Lee, J. Zhang, J. Wang, H. Han, Z. Yang, R. Fang, W. L. Tsai, H. Yang, E. Sag, R. Topaloglu, I. Aksentijevich, X. Yu, Q. Zhou, Deubiquitination of proteasome subunits by OTULIN regulates type I IFN production. *Sci. Adv.* **7**, eabi6794 (2021).
97. S. Davidson, C. H. Yu, A. Steiner, F. Ebstein, P. J. Baker, V. Jarur-Chamy, K. Hrovat Schaaale, P. Laohamonthonkul, K. Kong, D. J. Calleja, C. R. Harapas, K. R. Balka, J. Mitchell, J. T. Jackson, N. D. Geoghegan, F. Moghaddas, K. L. Rogers, K. D. Mayer-Barber, A. A. De Jesus, D. De Nardo, B. T. Kile, A. J. Sadler, M. C. Poli, E. Kruger, R. Goldbach Mansky, S. L. Masters, Protein kinase R is an innate immune sensor of proteotoxic stress via accumulation of cytoplasmic IL-24. *Sci. Immunol.* **7**, eabi6763 (2022).
98. K. Moriya, T. Nakano, Y. Honda, M. Tsumura, M. Ogishi, M. Sonoda, M. Nishitani-Isa, T. Uchida, M. Hibi, Y. Mizoguchi, M. Ishimura, K. Izawa, T. Asano, F. Kakuta, D. Abukawa, D. Rinchai, P. Zhang, N. Kambe, A. Bousfha, T. Yasumi, B. Boisson, A. Puel, J. L. Casanova, R. Nishikomori, S. Ohga, S. Okada, Y. Sasahara, S. Kure, Human RELA dominant-negative mutations underlie type I interferonopathy with autoinflammation and autoimmunity. *J. Exp. Med.* **220**, e20212276 (2023).
99. J. Wang, S. H. Basagoudanavar, X. Wang, E. Hopewell, R. Albrecht, A. Garcia-Sastre, S. Balachandran, A. A. Beg, NF-kappa B RelA subunit is crucial for early IFN-beta expression and resistance to RNA virus replication. *J. Immunol.* **185**, 1720–1729 (2010).
100. M. Iwanaszko, M. Kimmel, NF-kB and IRF pathways: Cross-regulation on target genes promoter level. *BMC Genomics* **16**, 307 (2015).

101. M. Czerkies, Z. Korwek, W. Prus, M. Kochanczyk, J. Jaruszewicz-Blonska, K. Tudelska, S. Blonski, M. Kimmel, A. R. Brasier, T. Lipniacki, Cell fate in antiviral response arises in the crosstalk of IRF, NF- κ B and JAK/STAT pathways. *Nat. Commun.* **9**, 493 (2018).
102. D. B. Stetson, J. S. Ko, T. Heidmann, R. Medzhitov, Trex1 prevents cell-intrinsic initiation of autoimmunity. *Cell* **134**, 587–598 (2008).
103. J. C. Hartner, C. R. Walkley, J. Lu, S. H. Orkin, ADAR1 is essential for the maintenance of hematopoiesis and suppression of interferon signaling. *Nat. Immunol.* **10**, 109–115 (2009).
104. M. Maurano, J. M. Snyder, C. Connelly, J. Henao-Mejia, C. Sidrauski, D. B. Stetson, Protein kinase R and the integrated stress response drive immunopathology caused by mutations in the RNA deaminase ADAR1. *Immunity* **54**, 1948–1960.e5 (2021).
105. H. E. Sonmez, C. Karaaslan, A. A. de Jesus, E. D. Batu, B. Anlar, B. Sozeri, Y. Bilginer, D. Karaguzel, D. Cagdas Ayvaz, I. Tezcan, R. Goldbach-Mansky, S. Ozen, A clinical score to guide in decision making for monogenic type I IFNopathies. *Pediatr. Res.* **87**, 745–752 (2020).
106. J. Wu, N. Dobbs, K. Yang, N. Yan, Interferon-independent activities of mammalian STING mediate antiviral response and tumor immune evasion. *Immunity* **53**, 115–126.e5 (2020).
107. L. H. Yamashiro, S. C. Wilson, H. M. Morrison, V. Karalis, J. J. Chung, K. J. Chen, H. S. Bateup, M. L. Szpara, A. Y. Lee, J. S. Cox, R. E. Vance, Interferon-independent STING signaling promotes resistance to HSV-1 in vivo. *Nat. Commun.* **11**, 3382 (2020).
108. S. Yum, M. Li, Y. Fang, Z. J. Chen, TBK1 recruitment to STING activates both IRF3 and NF- κ B that mediate immune defense against tumors and viral infections. *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2100225118 (2021).
109. J. D. Warner, R. A. Irizarry-Caro, B. G. Bennion, T. L. Ai, A. M. Smith, C. A. Miner, T. Sakai, V. K. Gonugunta, J. Wu, D. J. Platt, N. Yan, J. J. Miner, STING-associated vasculopathy develops independently of IRF3 in mice. *J. Exp. Med.* **214**, 3279–3292 (2017).
110. M. Motwani, S. Pawaria, J. Bernier, S. Moses, K. Henry, T. Fang, L. Burkly, A. Marshak-Rothstein, K. A. Fitzgerald, Hierarchy of clinical manifestations in SAVI N153S and V154M mouse models. *Proc. Natl. Acad. Sci. U.S.A.* **116**, 7941–7950 (2019).
111. G. R. Martin, K. Henare, C. Salazar, T. Scheidl-Yee, L. J. Eggen, P. P. Taylor, J. H. Kim, J. Podstawka, M. J. Fritzler, M. M. Kelly, B. G. Yipp, F. R. Jirik, Expression of a constitutively active human STING mutant in hematopoietic cells produces an Ifnar1-dependent vasculopathy in mice. *Life Sci. Alliance* **2**, e201800215 (2019).
112. J. Conlon, D. L. Burdette, S. Sharma, N. Bhat, M. Thompson, Z. Jiang, V. A. Rathinam, B. Monks, T. Jin, T. S. Xiao, S. N. Vogel, R. E. Vance, K. A. Fitzgerald, Mouse, but not human STING, binds and signals in response to the vascular disrupting agent 5,6-dimethylxanthenone-4-acetic acid. *J. Immunol.* **190**, 5216–5225 (2013).
113. T. Cavlar, T. Deimling, A. Ablasser, K. P. Hopfner, V. Hornung, Species-specific detection of the antiviral small-molecule compound CMA by STING. *EMBO J.* **32**, 1440–1450 (2013).
114. A. H. F. Andersen, R. Olesen, K. L. Jonsson, J. F. Hojen, C. Krapp, K. Mack, M. K. Thomsen, L. Ostergaard, M. Tolstrup, F. Dagnaes-Hansen, M. R. Jakobsen, P. W. Denton, cAIMP administration in humanized mice induces a chimerization-level-dependent STING response. *Immunology* **157**, 163–172 (2019).
115. W. Zhou, A. T. Whiteley, C. C. de Oliveira Mann, B. R. Morehouse, R. P. Nowak, E. S. Fischer, N. S. Gray, J. J. Mekalanos, P. J. Kranzusch, Structure of the human cGAS-DNA complex reveals enhanced control of immune surveillance. *Cell* **174**, 300–311.e11 (2018).
116. E. S. Pang, G. Daraj, K. R. Balka, D. De Nardo, C. Macri, H. Hochrein, K. A. Masterman, P. S. Tan, A. Shoppee, Z. Magill, N. Jahan, M. Baft, Y. Zhan, B. T. Kile, K. E. Lawlor, K. J. Radford, M. D. Wright, M. O'Keeffe, Discordance in STING-induced activation and cell death between mouse and human dendritic cell populations. *Front. Immunol.* **13**, 794776 (2022).
117. K. Kawane, H. Fukuyama, G. Kondoh, J. Takeda, Y. Ohsawa, Y. Uchiyama, S. Nagata, Requirement of DNase II for definitive erythropoiesis in the mouse fetal liver. *Science* **292**, 1546–1549 (2001).
118. H. Yoshida, Y. Okabe, K. Kawane, H. Fukuyama, S. Nagata, Lethal anemia caused by interferon-beta produced in mouse embryos carrying undigested DNA. *Nat. Immunol.* **6**, 49–56 (2005).
119. K. Kawane, M. Ohtani, K. Miwa, T. Kizawa, Y. Kanbara, Y. Yoshioka, H. Yoshikawa, S. Nagata, Chronic polyarthritis caused by mammalian DNA that escapes from degradation in macrophages. *Nature* **443**, 998–1002 (2006).
120. T. Li, S. Yum, M. Li, X. Chen, X. Zuo, Z. J. Chen, TBK1 recruitment to STING mediates autoinflammatory arthritis caused by defective DNA clearance. *J. Exp. Med.* **219**, e20211539 (2022).
121. K. Pestal, C. C. Funk, J. M. Snyder, N. D. Price, P. M. Treuting, D. B. Stetson, Isoforms of RNA-editing enzyme ADAR1 independently control nucleic acid sensor MDA5-driven autoimmunity and multi-organ development. *Immunity* **43**, 933–944 (2015).
122. M. A. Reijns, B. Rabe, R. E. Rigby, P. Mill, K. R. Astell, L. A. Lettice, S. Boyle, A. Leitch, M. Keighren, F. Kilanowski, P. S. Devenney, D. Sexton, G. Grimes, I. J. Holt, R. E. Hill, M. S. Taylor, K. A. Lawson, J. R. Dorin, A. P. Jackson, Enzymatic removal of ribonucleotides from DNA is essential for mammalian genome integrity and development. *Cell* **149**, 1008–1022 (2012).
123. J. Taft, M. Markson, D. Legarda, R. Patel, M. Chan, L. Malle, A. Richardson, C. Gruber, M. Martin-Fernandez, G. M. S. Mancini, J. A. M. van Laar, P. van Pelt, S. Buta, B. H. A. Wokke, I. K. D. Sabli, V. Sancho-Shimizu, P. P. Chavan, O. Schnappauf, R. Khubchandani, M. K. Cuceoglu, S. Ozen, D. L. Kastner, A. T. Ting, I. Aksentijevich, I. Hollink, D. Bogunovic, Human TBK1 deficiency leads to autoinflammation driven by TNF-induced cell death. *Cell* **184**, 4447–4463.e20 (2021).
124. T. Magg, T. Okano, L. M. Koenig, D. F. R. Boehmer, S. L. Schwartz, K. Inoue, J. Heimall, F. Licciardi, J. Ley-Zaporozhan, R. M. Ferdman, A. Caballero-Oteyza, E. N. Park, B. M. Calderon, D. Dey, H. Kanegane, K. Cho, D. Montin, K. Reiter, M. H. Albert, M. Rohlf, P. Gray, C. Walz, G. L. Conn, K. E. Sullivan, C. Klein, T. Morio, F. Hauck, Heterozygous OAS1 gain-of-function variants cause an autoinflammatory immunodeficiency. *Sci. Immunol.* **6**, eabf9564 (2021).
125. G. J. Brown, P. F. Canete, H. Wang, A. Medhavy, J. Bones, J. A. Roco, Y. He, Y. Qin, J. Cappello, J. I. Ellyard, K. Bassett, Q. Shen, G. Burgio, Y. Zhang, C. Turnbull, X. Meng, P. Wu, E. Cho, L. A. Miosge, T. D. Andrews, M. A. Field, D. Tvorogov, A. F. Lopez, J. J. Babon, C. A. Lopez, A. Gonzalez-Murillo, D. C. Garulo, V. Pascual, T. Levy, E. J. Mallack, D. G. Calame, T. Lotze, J. R. Lupski, H. Ding, T. R. Ullah, G. D. Walters, M. E. Koina, M. C. Cook, N. Shen, C. de Lucas Collantes, B. Corry, M. P. Gantier, V. Athanasopoulos, C. G. Vinuesa, TLR7 gain-of-function genetic variation causes human lupus. *Nature* **605**, 349–356 (2022).
126. C. David, M. Badonyi, R. Kechiche, A. Insalaco, M. Zecca, F. De Benedetti, S. Orcesi, L. Chiapparini, P. Comoli, S. Federici, M. Gattorno, M. Ginevri, E. Giorgio, V. Matteo, P. Moran-Alvarez, D. Politano, G. Prencipe, F. Sirchia, S. Volpi, C. Masson, G. I. Rice, M. L. Fremont, A. Lepelletier, J. A. Marsh, Y. J. Crow, Interface gain-of-function mutations in TLR7 cause systemic and neuro-inflammatory disease. *J. Clin. Immunol.* **44**, 60 (2024).
127. C. Wolf, E. L. Lim, M. Mokhtari, B. Kind, A. Odanic, E. Lara-Villacanas, S. Koss, S. Mages, K. Menzel, K. Engel, G. Duckers, B. Bernbeck, D. T. Schneider, K. Siepermann, T. Niehues, C. C. Goetzke, P. Durek, K. Minden, T. Dörner, A. Stittich, F. Szelinski, G. M. Guerra, M. Massoud, M. Bieringer, C. C. de Oliveira Mann, E. Beltran, T. Kallinich, M. F. Mashreghi, S. V. Schmidt, E. Latz, J. Klughammer, O. Majer, M. A. Lee-Kirsch, UNC93B1 variants underlie TLR7-dependent autoimmunity. *Sci. Immunol.* **9**, eadi9769 (2024).
128. H. Mishra, C. Schlack-Leigers, E. L. Lim, O. Thieck, T. Magg, J. Raedler, C. Wolf, C. Klein, H. Ewers, M. A. Lee-Kirsch, D. Meierhofer, F. Hauck, O. Majer, Disrupted degradative sorting of TLR7 is associated with human lupus. *Sci. Immunol.* **9**, eadi9575 (2024).
129. O. Majer, B. Liu, B. J. Woo, L. S. M. Kreuk, E. Van Dis, G. M. Barton, Release from UNC93B1 reinforces the compartmentalized activation of select TLRs. *Nature* **575**, 371–374 (2019).
130. O. Majer, B. Liu, L. S. M. Kreuk, N. Krogan, G. M. Barton, UNC93B1 recruits syntenin-1 to dampen TLR7 signalling and prevent autoimmunity. *Nature* **575**, 366–370 (2019).
131. M. Motwani, J. McGowan, J. Antonovitch, K. M. Gao, Z. Jiang, S. Sharma, G. A. Baltus, K. M. Nickerson, A. Marshak-Rothstein, K. A. Fitzgerald, cGAS-STING pathway does not promote autoimmunity in murine models of SLE. *Front. Immunol.* **12**, 605930 (2021).
132. G. I. Rice, S. Park, F. Gavazzi, L. A. Adang, L. A. Ayuk, L. Van Eyck, L. Seabra, C. Barrea, R. Battini, A. Belot, S. Berg, T. Billette de Villemeur, A. E. Bley, L. Blumkin, O. Boespflug-Tanguy, T. A. Briggs, E. Brimble, R. C. Dale, N. Darin, F. G. Debray, V. De Giorgis, J. Denecke, D. Doummar, G. D. A. Hagelsrum, D. Eleftheriou, M. Estienne, E. Fazzi, F. Feillet, J. Gallii, N. Hartog, J. Harvengt, B. Heron, D. Heron, D. A. Kelly, D. Lev, V. Levrat, J. H. Livingston, I. Marti, C. Mignot, F. Mochel, M. C. Nougues, I. Oppermann, B. Perez-Duenas, B. Popp, M. P. Rodero, D. Rodriguez, V. Saletti, C. Sharpe, D. Tonduti, G. Vadlamani, K. Van Haren, M. T. Vila, J. Vogt, E. Wassmer, A. Wiedemann, C. J. Wilson, A. Zerem, C. Zweier, S. M. Zuberi, S. Orcesi, A. L. Vanderver, S. Hur, Y. J. Crow, Genetic and phenotypic spectrum associated with IFIH1 gain-of-function. *Hum. Mutat.* **41**, 837–849 (2020).
133. C. J. A. Duncan, S. Hambleton, Human disease phenotypes associated with loss and gain of function mutations in STAT2: Viral susceptibility and type I interferonopathy. *J. Clin. Immunol.* **41**, 1446–1456 (2021).
134. J. Toubiana, S. Okada, J. Hiller, M. Oleastro, M. Lagos Gomez, J. C. Aldave Becerra, M. Ouachee-Chardin, F. Fouyssac, K. M. Girisha, A. Etzioni, J. Van Montfrans, Y. Camcioglu, L. A. Kerns, B. Belohradsky, S. Blanche, A. Bousfiha, C. Rodriguez-Gallego, I. Meyts, K. Kisand, J. Reichenbach, E. D. Renner, S. Rosenzweig, B. Grimbacher, F. L. van de Veerdonk, C. Traidl-Hoffmann, C. Picard, L. Marodi, T. Morio, M. Kobayashi, D. Lilic, J. D. Milner, S. Holland, J. L. Casanova, A. Puel, S. G.-o.-F. S. G., Heterozygous STAT1 gain-of-function mutations underlie an unexpectedly broad clinical phenotype. *Blood* **127**, 3154–3164 (2016).
135. Q. Zhang, P. Bastard, C. H. G. Effort, A. Cobat, J. L. Casanova, Human genetic and immunological determinants of critical COVID-19 pneumonia. *Nature* **603**, 587–598 (2022).
136. F. Gothe, J. Stremenova Spegarova, C. F. Hatton, H. Griffin, T. Sargent, S. A. Cowley, W. James, A. Roppelt, A. Shcherbina, F. Hauck, H. T. Reyburn, C. J. A. Duncan, S. Hambleton, Aberrant inflammatory responses to type I interferon in STAT2 or IRF9 deficiency. *J. Allergy Clin. Immunol.* **150**, 955–964.e16 (2022).
137. F. Gothe, C. F. Hatton, L. Truong, Z. Klimova, V. Kanderova, M. Fejtikova, A. Grainger, V. Bigley, J. Perthen, D. Mitra, A. Janda, E. Fronkova, D. Moravcikova, S. Hambleton, C. J. A. Duncan, A novel case of homozygous interferon alpha/beta receptor alpha chain (IFNAR1) deficiency with hemophagocytic lymphohistiocytosis. *Clin. Infect. Dis.* **74**, 136–139 (2022).

138. A. Schmidt, S. Peters, A. Knaus, H. Sabir, F. Hamsen, C. Maj, J. Fazaal, S. Sivalingam, O. Savchenko, A. Mantri, D. Holzinger, U. Neudorf, A. Muller, K. U. Ludwig, P. M. Krawitz, H. Engels, M. M. Nothen, S. Bagci, TBK1 and TNFRSF13B mutations and an autoinflammatory disease in a child with lethal COVID-19. *NPJ Genom. Med.* **6**, 55 (2021).
139. R. G. Saldanha, K. R. Balka, S. Davidson, B. K. Wainstein, M. Wong, R. Macintosh, C. K. C. Loo, M. A. Weber, V. Kamath, CIRCA, ADHY, F. Moghaddas, D. De Nardo, P. E. Gray, S. L. Masters, A mutation outside the dimerization domain causing atypical STING-associated vasculopathy with onset in infancy. *Front. Immunol.* **9**, 1535 (2018).
140. T. Kato, M. Yamamoto, Y. Honda, T. Orimo, I. Sasaki, K. Murakami, H. Hemmi, Y. Fukuda-Ohta, K. Isono, S. Takayama, H. Nakamura, Y. Otsuki, T. Miyamoto, J. Takita, T. Yasumi, R. Nishikomori, T. Matsubayashi, K. Izawa, T. Kaisho, Augmentation of stimulator of interferon genes-induced type I interferon production in COPA syndrome. *Arthritis Rheumatol.* **73**, 2105–2115 (2021).
141. M. Zeleznik, A. Soltirovska Salamon, M. Debeljak, A. Goropevsek, N. Sustar, D. Kljucevsek, A. Ihan, T. Avcin, Case report: *Pneumocystis jirovecii* pneumonia in a severe case of Aicardi-Goutieres syndrome with an IFIH1 gain-of-function mutation mimicking combined immunodeficiency. *Front. Immunol.* **13**, 1033513 (2022).
142. Y. Kida, T. Doi, Y. Imanaka, V. J. Amatya, N. Shime, S. Okada, Lethal interstitial lung disease associated with a gain-of-function mutation in IFIH1. *J. Clin. Immunol.* **43**, 1143–1146 (2023).
143. T. T. Greene, E. I. Zuniga, Type I interferon induction and exhaustion during viral infection: Plasmacytoid dendritic cells and emerging COVID-19 findings. *Viruses* **13**, 1839 (2021).
144. P. M. Odorizzi, E. J. Wherry, Immunology. An interferon paradox. *Science* **340**, 155–156 (2013).
145. L. Moreira-Teixeira, K. Mayer-Barber, A. Sher, A. O'Garra, Type I interferons in tuberculosis: Foe and occasionally friend. *J. Exp. Med.* **215**, 1273–1285 (2018).
146. V. Sancho-Shimizu, R. Perez de Diego, L. Lorenzo, R. Halwani, A. Alangari, E. Israelsson, S. Fabrega, A. Cardon, J. Maluenda, M. Tatematsu, F. Mahvelati, M. Herman, M. Ciancanelli, Y. Guo, Z. AlSum, N. Alkhamis, A. S. Al-Makadma, A. Ghadiri, S. Bouchier, S. Plancoulaine, C. Picard, F. Rozenberg, M. Tardieu, P. Lebon, E. Jouanguy, N. Rezaei, T. Seya, M. Matsumoto, D. Chaussabel, A. Puel, S. Y. Zhang, L. Abel, S. Al-Muhsen, J. L. Casanova, Herpes simplex encephalitis in children with autosomal recessive and dominant TRIF deficiency. *J. Clin. Invest.* **121**, 4889–4902 (2011).
147. R. Perez de Diego, V. Sancho-Shimizu, L. Lorenzo, A. Puel, S. Plancoulaine, C. Picard, M. Herman, A. Cardon, A. Durandy, J. Bustamante, S. Vallabhapurapu, J. Bravo, K. Warnatz, Y. Chaix, F. Cascarrigny, P. Lebon, F. Rozenberg, M. Karin, M. Tardieu, S. Al-Muhsen, E. Jouanguy, S. Y. Zhang, L. Abel, J. L. Casanova, Human TRAF3 adaptor molecule deficiency leads to impaired Toll-like receptor 3 response and susceptibility to herpes simplex encephalitis. *Immunity* **33**, 400–411 (2010).
148. A. P. Michaud, N. M. Bauman, D. K. Burke, J. M. Manaligod, R. J. Smith, Spastic diplegia and other motor disturbances in infants receiving interferon-alpha. *Laryngoscope* **114**, 1231–1236 (2004).
149. H. Worle, E. Maass, B. Kohler, J. Treuner, Interferon alpha-2a therapy in haemangiomas of infancy: Spastic diplegia as a severe complication. *Eur. J. Pediatr.* **158**, 344 (1999).
150. C. F. Barlow, C. J. Priebe, J. B. Mulliken, P. D. Barnes, D. Mac Donald, J. Folkman, R. A. Ezekowitz, Spastic diplegia as a complication of interferon Alfa-2a treatment of hemangiomas of infancy. *J. Pediatr.* **132**, 527–530 (1998).
151. T. Goldmann, N. Zeller, J. Raasch, K. Kierdorf, K. Frenzel, L. Ketscher, A. Basters, O. Staszewski, S. M. Brendecke, A. Spiess, T. L. Tay, C. Kreutz, J. Timmer, G. M. Mancini, T. Blank, G. Fritz, K. Biber, R. Lang, D. Moser, D. Merkler, M. Heikenwalder, K. P. Knobloch, M. Prinz, USP18 lack in microglia causes destructive interferonopathy of the mouse brain. *EMBO J.* **34**, 1612–1629 (2015).
152. S. R. Paludan, T. Pradeu, S. L. Masters, T. H. Mogensen, Constitutive immune mechanisms: Mediators of host defence and immune regulation. *Nat. Rev. Immunol.* **21**, 137–150 (2021).
153. S. Y. Zhang, N. E. Clark, C. A. Freije, E. Pauwels, A. J. Taggart, S. Okada, H. Mandel, P. Garcia, M. J. Ciancanelli, A. Biran, F. G. Lafaille, M. Tsumura, A. Cobat, J. Luo, S. Volpi, B. Zimmer, S. Sakata, A. Dinis, O. Ohara, E. J. Garcia Reino, K. Dobbs, M. Hasek, S. P. Holloway, K. McCammon, S. A. Hussong, N. DeRosa, C. E. Van Skike, A. Katolik, L. Lorenzo, M. Hyodo, E. Faria, R. Halwani, R. Fukuhara, G. A. Smith, V. Galvan, M. J. Damha, S. Al-Muhsen, Y. Itan, J. D. Boeke, L. D. Notarangelo, L. Studer, M. Kobayashi, L. Diogo, W. G. Fairbrother, L. Abel, B. R. Rosenberg, P. J. Hart, A. Etzioni, J. L. Casanova, Inborn errors of RNA lariat metabolism in humans with brainstem viral infection. *Cell* **172**, 952–965.e18 (2018).
154. F. G. Lafaille, O. Harschnitz, Y. S. Lee, P. Zhang, M. L. Hasek, G. Kerner, Y. Itan, O. Ewaleifoh, F. Rapaport, T. M. Carlile, M. E. Carter-Timothe, D. Paquet, K. Dobbs, B. Zimmer, D. Gao, M. F. Rojas-Duran, D. Kwart, V. Rattina, M. J. Ciancanelli, J. L. McAlpine, L. Lorenzo, S. Bouchier, F. Rozenberg, R. Halwani, B. Henry, N. Amenouz, Z. Alsum, L. Marques, J. A. Church, S. Al-Muhsen, M. Tardieu, A. A. Bousfiha, S. R. Paludan, T. H. Mogensen, L. Quintana-Murci, M. Tessier-Lavigne, G. A. Smith, L. D. Notarangelo, L. Studer, W. Gilbert, L. Abel, J. L. Casanova, S. Y. Zhang, Human SNORA31 variations impair cortical neuron-intrinsic immunity to HSV-1 and underlie herpes simplex encephalitis. *Nat. Med.* **25**, 1873–1884 (2019).
155. A. S. Hait, D. Olagnier, V. Sancho-Shimizu, K. A. Skipper, M. Helleberg, S. M. Larsen, C. Bodda, L. I. Moldovan, F. Ren, N. S. Brinck Andersen, M. M. Thomsen, M. R. Freytag, S. Darmalinggam, I. Parkes, D. D. Kadekar, S. H. Rahbek, D. van der Horst, L. S. Kristensen, K. Eriksson, J. Kjems, S. Mostowy, M. Christiansen, J. G. Mikkelsen, C. T. Brandt, S. R. Paludan, T. H. Mogensen, Defects in LC3B2 and ATG4A underlie HSV2 meningitis and reveal a critical role for autophagy in antiviral defense in humans. *Sci. Immunol.* **5**, eabc2691 (2020).
156. D. J. Gough, N. L. Messina, C. J. Clarke, R. W. Johnstone, D. E. Levy, Constitutive type I interferon modulates homeostatic balance through tonic signaling. *Immunity* **36**, 166–174 (2012).
157. P. Deb, J. Dai, S. Singh, E. Kalyoussef, P. Fitzgerald-Bocarsly, Triggering of the cGAS-STING pathway in human plasmacytoid dendritic cells inhibits TLR9-mediated IFN production. *J. Immunol.* **205**, 223–236 (2020).
158. D. Jeltima, K. Abbott, N. Yan, STING trafficking as a new dimension of immune signaling. *J. Exp. Med.* **220**, e20220990 (2023).
159. R. Channappanavar, A. R. Fehr, J. Zheng, C. Wohlford-Lenane, J. E. Abrahante, M. Mack, R. Sompallae, P. B. McCray Jr., D. K. Meyerholz, S. Perlman, IFN-I response timing relative to virus replication determines MERS coronavirus infection outcomes. *J. Clin. Invest.* **129**, 3625–3639 (2019).
160. C. Urban, H. Welsch, K. Heine, S. Wust, D. A. Haas, C. Dachert, A. Pandey, A. Pichlmair, M. Binder, Persistent innate immune stimulation results in IRF3-mediated but caspase-independent cytostasis. *Viruses* **12**, 635 (2020).
161. J. Loske, J. Rohmel, S. Lukassen, S. Stricker, V. G. Magalhaes, J. Liebig, R. L. Chua, L. Thurmman, M. Messingschlager, A. Seegebarth, B. Timmermann, S. Klages, M. Ralser, B. Sawitzki, L. E. Sander, V. M. Corman, C. Conrad, S. Laudi, M. Binder, S. Trump, R. Eils, M. A. Mall, I. Lehmann, Pre-activated antiviral innate immunity in the upper airways controls early SARS-CoV-2 infection in children. *Nat. Biotechnol.* **40**, 319–324 (2022).
162. S. S. Burkart, D. Schweinich, J. Frankish, C. Sparr, S. Wust, C. Urban, M. Merlo, V. G. Magalhaes, A. Piras, A. Pichlmair, J. Willemsen, L. Kaderali, M. Binder, High-resolution kinetic characterization of the RIG-I-signaling pathway and the antiviral response. *Life Sci. Alliance* **6**, e202302059 (2023).
163. C. Gruber, D. Bogunovic, Incomplete penetrance in primary immunodeficiency: A skeleton in the closet. *Hum. Genet.* **139**, 745–757 (2020).
164. A. Lepellet, M. J. Martin-Niclos, M. Le Bihan, J. A. Marsh, C. Ugenti, G. I. Rice, V. Bondet, D. Duffy, J. Hertzog, J. Rehwinkel, S. Amselem, S. Boulisfane-El Khalifi, M. Brennan, E. Carter, L. Chatenoud, S. Chhun, A. C. L'Hermine, M. Depp, M. Legendre, K. J. Mackenzie, J. Marey, C. McDougall, K. J. McKenzie, T. J. Molina, B. Neven, L. Seabra, C. Thumerelle, M. Wislez, N. Nathan, N. Manel, Y. J. Crow, M. L. Fremont, Mutations in COPA lead to abnormal trafficking of STING to the Golgi and interferon signaling. *J. Exp. Med.* **217**, e20200600 (2020).
165. Y. J. Crow, AGS group, Clinical non-penetrance associated with biallelic mutations in the RNase H2 complex. *J. Clin. Immunol.* **43**, 706–708 (2023).
166. L. Lodi, I. Melki, V. Bondet, L. Seabra, G. I. Rice, E. Carter, A. Lepellet, M. J. Martin-Niclos, B. Al Adba, B. Bader-Meunier, M. Barth, T. Blauwblomme, C. Bodemer, O. Boespflug-Tanguy, R. C. Dale, I. Desguerre, C. Ducrocq, F. Dulieu, C. Dumaine, P. Ellul, A. Hadchouel, V. Hentgen, M. Hie, M. Hully, E. Jeziorski, R. Levy, F. Mochel, S. Orcesi, S. Passemard, M. Pouletty, P. Quartier, F. Renaldo, R. Seidl, J. Shetty, B. Neven, S. Blanche, D. Duffy, Y. J. Crow, M. L. Fremont, Differential expression of interferon-alpha protein provides clues to tissue specificity across type I interferonopathies. *J. Clin. Immunol.* **41**, 603–609 (2021).
167. J. Maelfait, J. Rehwinkel, The Z-nucleic acid sensor ZBP1 in health and disease. *J. Exp. Med.* **220**, e20221156 (2023).
168. M. M. Painter, J. H. Morrison, L. J. Zoecklein, T. A. Rinkoski, J. O. Watzlawik, L. M. Papke, A. E. Warrington, A. J. Bieber, W. E. Matchett, K. L. Turkowski, E. M. Poeschla, M. Rodriguez, Antiviral protection via RdRP-mediated stable activation of innate immunity. *PLOS Pathog.* **11**, e1005311 (2015).
169. T. A. Briggs, G. I. Rice, S. Daly, J. Urquhart, H. Gornall, B. Bader-Meunier, K. Baskar, S. Baskar, V. Baudouin, M. W. Beresford, G. C. Black, R. J. Dearman, F. de Zegher, E. S. Foster, C. Frances, A. R. Hayman, E. Hilton, C. Job-Deslandre, M. L. L. Kulkarni, M. Le Merrer, A. Linglart, S. C. Lovell, K. Maurer, L. Musset, V. Navarro, C. Picard, A. Puel, F. Rieux-Laucat, C. M. Roifman, S. Scholl-Burgi, N. Smith, M. Szykiewicz, A. Wiedeman, C. Wouters, L. A. Zeef, J. L. Casanova, K. B. Elkon, A. Janclick, P. Lebon, Y. J. Crow, Tartrate-resistant acid phosphatase deficiency causes a bone dysplasia with autoimmunity and a type I interferon expression signature. *Nat. Genet.* **43**, 127–131 (2011).
170. O. Omarjee, C. Picard, C. Frachette, M. Moreeys, F. Rieux-Laucat, P. Soulas-Sprauel, S. Viel, J. C. Lega, B. Bader-Meunier, T. Walzer, A. L. Mathieu, R. Cimaz, A. Belot, Monogenic lupus: Dissecting heterogeneity. *Autoimmun. Rev.* **18**, 102361 (2019).
171. L. M. Booshehri, H. M. Hoffman, CAPS and NLRP3. *J. Clin. Immunol.* **39**, 277–286 (2019).
172. P. D. Monk, R. J. Marsden, V. J. Tear, J. Brookes, T. N. Batten, M. Mankowski, F. J. Gabbay, D. E. Davies, S. T. Holgate, L. P. Ho, T. Clark, R. Djukanovic, T. M. A. Wilkinson, Inhaled Interferon Beta COVID-19 Study Group, Safety and efficacy of inhaled nebulised

- interferon beta-1a (SNG001) for treatment of SARS-CoV-2 infection: A randomised, double-blind, placebo-controlled, phase 2 trial. *Lancet Respir. Med.* **9**, 196–206 (2021).
173. T. Le Voyer, S. Sakata, M. Tsumura, T. Khan, A. Esteve-Sole, B. K. Al-Saud, H. E. Gungor, P. Taur, V. Jeanne-Julien, M. Christiansen, L. M. Kohler, G. E. ElGhazali, J. Rosain, S. Nishimura, F. Sakura, M. Bouaziz, C. Oleaga-Quintas, A. Nieto-Patlan, A. Deya-Martinez, Y. Altuner Torun, A. L. Neehus, M. Roynard, S. E. Bozdemir, N. Al Kaabi, M. Al Hassani, I. Mersiyanova, F. Rozenberg, C. Speckmann, I. Hainmann, F. Hauck, M. H. Alzahrani, S. H. Alhajjar, S. Al-Muhsen, T. Cole, R. Fuleihan, P. D. Arkwright, R. Badolato, L. Alsina, L. Abel, M. Desai, H. Al-Mousa, A. Shcherbina, N. Marr, S. Boisson-Dupuis, J. L. Casanova, S. Okada, J. Bustamante, Genetic, immunological, and clinical features of 32 patients with autosomal recessive STAT1 deficiency. *J. Immunol.* **207**, 133–152 (2021).
 174. M. L. Fremond, M. P. Roderio, N. Jeremiah, A. Belot, E. Jeziorski, D. Duffy, D. Bessis, G. Cros, G. I. Rice, B. Charbit, A. Hulin, N. Khoudour, C. M. Caballero, C. Bodemer, M. Fabre, L. Berteloot, M. Le Bourgeois, P. Reix, T. Walzer, D. Moshous, S. Blanche, A. Fischer, B. Bader-Meunier, F. Rieux-Laucat, Y. J. Crow, B. Neven, Efficacy of the Janus kinase 1/2 inhibitor ruxolitinib in the treatment of vasculopathy associated with TMEM173-activating mutations in 3 children. *J. Allergy Clin. Immunol.* **138**, 1752–1755 (2016).
 175. Y. Liu, A. A. Jesus, B. Marrero, D. Yang, S. E. Ramsey, G. A. M. Sanchez, K. Tenbrock, H. Wittkowski, O. Y. Jones, H. S. Kuehn, C. R. Lee, M. A. DiMattia, E. W. Cowen, B. Gonzalez, I. Palmer, J. J. DiGiovanna, A. Biancotto, H. Kim, W. L. Tsai, A. M. Trier, Y. Huang, D. L. Stone, S. Hill, H. J. Kim, C. St Hilaire, S. Gurprasad, N. Plass, D. Chapelle, I. Horkayne-Szakaly, D. Foell, A. Barysenka, F. Candotti, S. M. Holland, J. D. Hughes, H. Mehmet, A. C. Issekutz, M. Raffeld, J. McElwee, J. R. Fontana, C. P. Minniti, S. Moir, D. L. Kastner, M. Gadina, A. C. Steven, P. T. Wingfield, S. R. Brooks, S. D. Rosenzweig, T. A. Fleisher, Z. Deng, M. Boehm, A. S. Paller, R. Goldbach-Mansky, Activated STING in a vascular and pulmonary syndrome. *N. Engl. J. Med.* **371**, 507–518 (2014).
 176. Y. Liu, Y. Ramot, A. Torrello, A. S. Paller, N. Si, S. Babay, P. W. Kim, A. Sheikh, C. C. Lee, Y. Chen, A. Vera, X. Zhang, R. Goldbach-Mansky, A. Zlotogorski, Mutations in proteasome subunit β type 8 cause chronic atypical neutrophilic dermatosis with lipodystrophy and elevated temperature with evidence of genetic and phenotypic heterogeneity. *Arthritis Rheum.* **64**, 895–907 (2012).
 177. G. A. M. Sanchez, A. Reinhardt, S. Ramsey, H. Wittkowski, P. J. Hashkes, Y. Berkun, S. Schalm, S. Murias, J. A. Dare, D. Brown, D. L. Stone, L. Gao, T. Klausmeier, D. Foell, A. A. de Jesus, D. C. Chapelle, H. Kim, S. Dill, R. A. Colbert, L. Failla, B. Kost, M. O'Brien, J. C. Reynolds, L. R. Folio, K. R. Calvo, S. M. Paul, N. Weir, A. Brofferio, A. Soldatos, A. Biancotto, E. W. Cowen, J. J. Digiovanna, M. Gadina, A. J. Lipton, C. Hadigan, S. M. Holland, J. Fontana, A. S. Alawad, R. J. Brown, K. I. Rother, T. Heller, K. M. Brooks, P. Kumar, S. R. Brooks, M. Waldman, H. K. Singh, V. Nickleit, M. Silk, A. Prakash, J. M. Janes, S. Ozen, P. G. Wakim, P. L. Brogan, W. L. Macias, R. Goldbach-Mansky, JAK1/2 inhibition with baricitinib in the treatment of autoinflammatory interferonopathies. *J. Clin. Invest.* **128**, 3041–3052 (2018).
 178. A. Vanderver, L. Adang, F. Gavazzi, K. McDonald, G. Helman, D. B. Frank, N. Jaffe, S. W. Yum, A. Collins, S. R. Keller, P. Lebon, J. F. Meritet, J. Rhee, A. Takanoashi, T. Armangué, N. Ulrick, O. Sherbini, J. Koh, K. Peer, C. Besnier, C. Scher, K. Boyle, H. Dubbs, J. Kramer-Golinkoff, A. Pizzino, S. Woidill, J. Shults, Janus kinase inhibition in the Aicardi-Goutières syndrome. *N. Engl. J. Med.* **383**, 986–989 (2020).
 179. R. Goldbach-Mansky, N. J. Dailey, S. W. Canna, A. Gelabert, J. Jones, B. I. Rubin, H. J. Kim, C. Brewer, C. Zalewski, E. Wiggs, S. Hill, M. L. Turner, B. I. Karp, I. Aksentijevich, F. Pucino, S. R. Penzak, M. H. Haverkamp, L. Stein, B. S. Adams, T. L. Moore, R. C. Fuhlbrigge, B. Shaham, J. N. Jarvis, K. O'Neil, R. K. Vehe, L. O. Beitz, G. Gardner, W. P. Hannan, R. W. Warren, W. Horn, J. L. Cole, S. M. Paul, P. N. Hawkins, T. H. Pham, C. Snyder, R. A. Wesley, S. C. Hoffmann, S. M. Holland, J. A. Butman, D. L. Kastner, Neonatal-onset multisystem inflammatory disease responsive to interleukin-1beta inhibition. *N. Engl. J. Med.* **355**, 581–592 (2006).
 180. B. Neven, B. Al Adba, M. Hully, I. Desguerre, C. Pressiat, N. Bodaert, D. Duffy, G. I. Rice, L. Seabra, M. L. Fremond, S. Blanche, Y. J. Crow, JAK inhibition in the Aicardi-Goutières syndrome. *N. Engl. J. Med.* **383**, 2190–2191 (2020).
 181. M. L. Fremond, M. Hully, B. Fournier, R. Barrois, R. Levy, M. Aubart, M. Castelle, D. Chaballier, C. Gins, E. Sarda, B. Al Adba, S. Couderc, C. D. Almeida, C. M. Berat, C. Durrleman, C. Espil, L. Lambert, C. Meni, M. Perivier, P. Pillet, L. Polivka, M. Schiff, C. Todosi, F. Uettwiller, A. Lepelley, G. I. Rice, L. Seabra, S. Sanquer, A. Hulin, C. Pressiat, L. Goldwirt, V. BonDET, D. Duffy, D. Moshous, B. Bader-Meunier, C. Bodemer, F. Robin-Renaldo, N. Bodaert, S. Blanche, I. Desguerre, Y. J. Crow, B. Neven, JAK inhibition in Aicardi-Goutières syndrome: A monocentric multidisciplinary real-world approach study. *J. Clin. Immunol.* **43**, 1436–1447 (2023).
 182. B. Viengkhou, C. Hong, C. Mazur, S. Damle, N. B. Gallo, T. C. Fang, K. Henry, I. L. Campbell, F. Kamme, M. J. Hofer, Interferon- α receptor antisense oligonucleotides reduce neuroinflammation and neuropathology in a mouse model of cerebral interferonopathy. *J. Clin. Invest.* **134**, e169562 (2024).
 183. J. Taft, D. Bogunovic, The Goldilocks Zone of type I IFNs: Lessons from human genetics. *J. Immunol.* **201**, 3479–3485 (2018).
 184. G. I. Rice, G. M. Forte, M. Szykiewicz, D. S. Chase, A. Aeby, M. S. Abdel-Hamid, S. Ackroyd, R. Allcock, K. M. Bailey, U. Balottin, C. Barnerias, G. Bernard, C. Bodemer, M. P. Botella, C. Cereda, K. E. Chandler, L. Dabydeen, R. C. Dale, C. De Laet, C. G. De Goede, M. Del Toro, L. Effat, N. N. Enamorado, E. Fazzi, B. Gener, M. Haldre, J. P. Lin, J. H. Livingston, C. M. Lourenco, W. Marques Jr., P. Oades, P. Peterson, M. Rasmussen, A. Roubertie, J. L. Schmidt, S. A. Shalev, R. Simon, R. Spiegel, K. J. Swoboda, S. A. Temtamy, G. Vassallo, C. N. Vilain, J. Vogt, V. Wermenbol, W. P. Whitehouse, D. Soler, I. Olivieri, S. Orcesi, M. S. Aglan, M. S. Zaki, G. M. Abdel-Salam, A. Vanderver, K. Kisand, F. Rozenberg, P. Lebon, Y. J. Crow, Assessment of interferon-related biomarkers in Aicardi-Goutières syndrome associated with mutations in TREX1, RNASEH2A, RNASEH2B, RNASEH2C, SAMHD1, and ADAR: A case-control study. *Lancet Neurol.* **12**, 1159–1169 (2013).
 185. H. Kim, A. A. de Jesus, S. R. Brooks, Y. Liu, Y. Huang, R. VanTries, G. A. Montealegre Sanchez, Y. Rotman, M. Gadina, R. Goldbach-Mansky, Development of a validated interferon score using NanoString technology. *J. Interferon Cytokine Res.* **38**, 171–185 (2018).
 186. J. W. Schoggins, Interferon-stimulated genes: What do they all do? *Annu. Rev. Virol.* **6**, 567–584 (2019).
 187. A. E. Shaw, J. Hughes, Q. Gu, A. Behdenna, J. B. Singer, T. Dennis, R. J. Orton, M. Varela, R. J. Gifford, S. J. Wilson, M. Palmarini, Fundamental properties of the mammalian innate immune system revealed by multispecies comparison of type I interferon responses. *PLOS Biol.* **15**, e2004086 (2017).

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